2 Introduction to OpenOffice



SNAP RECAP

1. What do you understand by the term open source software?

   .....

   .....

2. Which software program(s) do you generally use for editing a document or creating a presentation?

   .....

   .....

LEARNING OBJECTIVES



You will learn about:

- various OpenOffice applications
- introduction to OpenOffice Writer
- main components of OpenOffice Writer window
- File, Edit and Format menus in OpenOffice Writer
- introduction to OpenOffice Impress
- creating, modifying and running a presentation in OpenOffice Impress

Introduction

OpenOffice is a collection of application software just like Microsoft Office. It is compatible with many operating systems like Windows, Linux and MacOS. **Sun Microsystems** released the first version of OpenOffice 1.0 but it was taken over by **Apache Foundation** in 2011. The software was renamed as **Apache OpenOffice**.

OpenOffice is an open source software application, hence its source code is available and can be freely downloaded from openoffice.org. It consists of six products as given in Table 2.1.

Writer

Calc

Impress

Math

Draw

Base

OpenOffice

OpenOffice Wri
document. It is
The extension o

It can be used t

- documents
- brochures a
- newsletter w

- Data types
- Primary key
- Changing the view
- Forms
- Reports

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- OpenOffice Writer
- OpenOffice Impress

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- OpenOffice Calc
- OpenOffice Base

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- Frames
- Frameset tag <FRAMESET>
- Frame tag <FRAME>
- Nested frameset

Main Components of OpenOffice Writer Window

To open OpenOffice Writer, click on **Start** button ⟹ **OpenOffice** ⟹ **OpenOffice Writer**. The main components of OpenOffice Writer window are shown in Fig. 2.1.

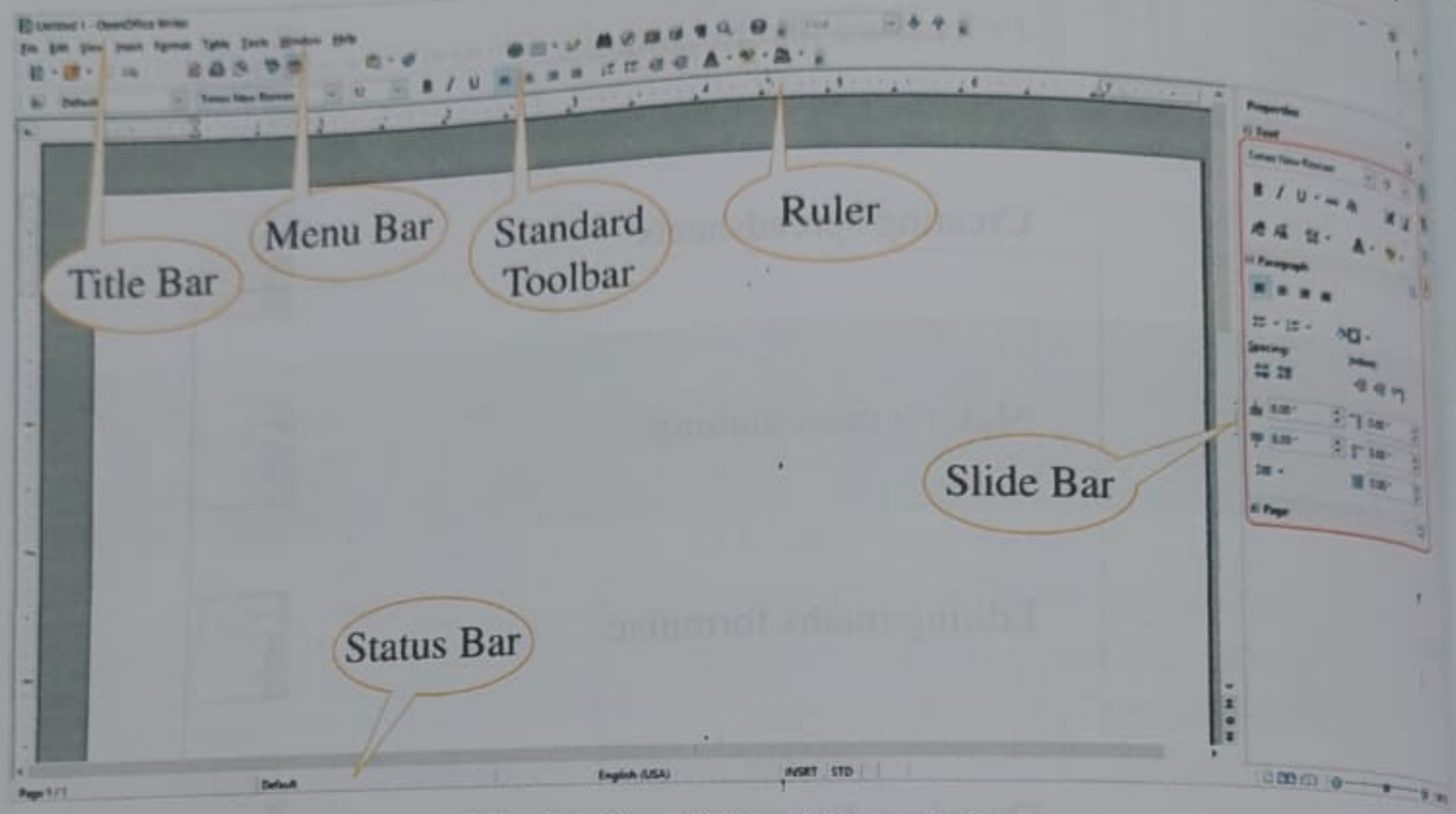


Fig. 2.1 *OpenOffice Writer window*

Title Bar

The bar on the top of the application window with document name Untitled 1 and program name OpenOffice Writer is known as the title bar.

Menu Bar

The menu bar is made up of many drop down menu tabs like File, Edit, View, Insert, Format, Table, Tools, Window and Help.

Standard Toolbar

It includes tools for commonly used tasks such as open, save, copy, cut, paste, etc.

Ruler

Both vertical and horizontal rulers are available to show dimensions, tabs, columns, margins of the page, etc.

Slide Bar

The slide bar contains frequently used tools grouped in a deck.

Status Bar

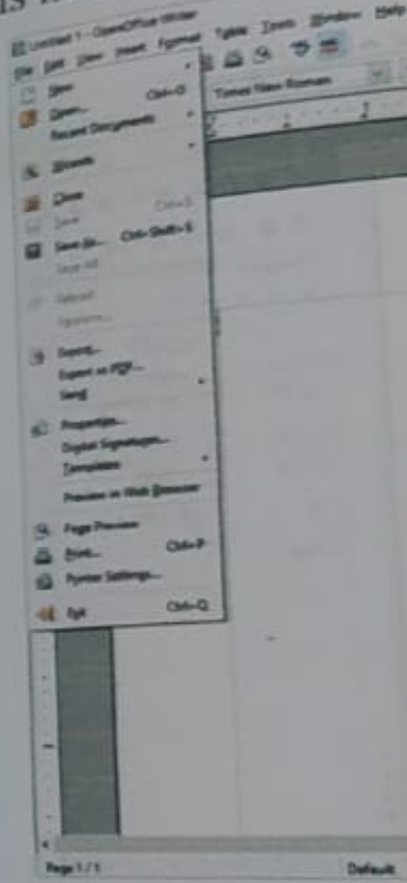
It displays information
insert or overwrite mod

Menus in the Oper

The menu bar in Open
Edit, View, Insert, For
these menus in detail.

File Menu

The File menu in Ope
this menu are discusse



New: Creates a new

Open...: Opens an

Close: Closes a file

Save: Saves the cha

Save As...: Saves a
name.

Save All: Saves all

Status Bar

It displays information about the current page, current template, zoom percentage, insert or overwrite mode, selection mode, unsaved changes and digital signature.

Menus in the OpenOffice Writer

The menu bar in OpenOffice Writer consists of various drop-down menu tabs like File, Edit, View, Insert, Format, Table, Tools, Window and Help. Let us discuss some of these menus in detail.

File Menu

The File menu in OpenOffice Writer is shown in Fig. 2.2. Various options available in this menu are discussed below:



Fig. 2.2 *File menu in OpenOffice Writer*

New: Creates a new file.

Open...: Opens an already created file on our computer.

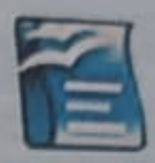
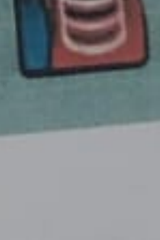
Close: Closes a file.

Save: Saves the changes in an already created file.

Save As...: Saves a new file for the first time or saves an existing file with a different name.

Save All: Saves all open files.

Table 2.1 *OpenOffice applications*

OpenOffice Application	Used for	Logo
Writer	Processing text documents	
Calc	Creating spreadsheets	
Impress	Making presentations	
Math	Editing maths formulae	
Draw	Drawing diagrams	
Base	Creating tables and database	

OpenOffice Writer

OpenOffice Writer is a word processor used to create, edit, format and print a document. It is just like MS Word with almost similar features but different layout. The extension of the file created in OpenOffice Writer is .odt.

It can be used to create and edit:

- documents with text, images and charts
- brochures and invitations
- newsletter with multiple columns.

Paste Special...: It pastes the formatted or unformatted text in a document.

Select All: It selects the complete document.

Find & Replace...: It searches for a selected word and can replace the searched word with a new word.



ACTIVITY

Launch the OpenOffice Writer using the Start button of your computer. Write a letter to a friend or a cousin living in some other city, asking them about their well-being. Try using different commands on the text that you have learnt so far.

TRY THIS



Dragging and dropping a selected text is equivalent to cut and paste in the same or another document.

- Select the text you wish to move.
- Place the mouse pointer anywhere on the selected text without clicking.
- Left click and drag the selected text to the new location.
- During this process, the mouse pointer changes to a box with a small white arrow over it, indicating that you are dragging text.
- When you reach the new location, release the mouse button to drop the text.

Format Menu

The Format menu in OpenOffice Writer is shown in Fig. 2.4. Various options available in this menu are discussed below:

Character...: The formatting related to the character like Font, Font Effects, Position,

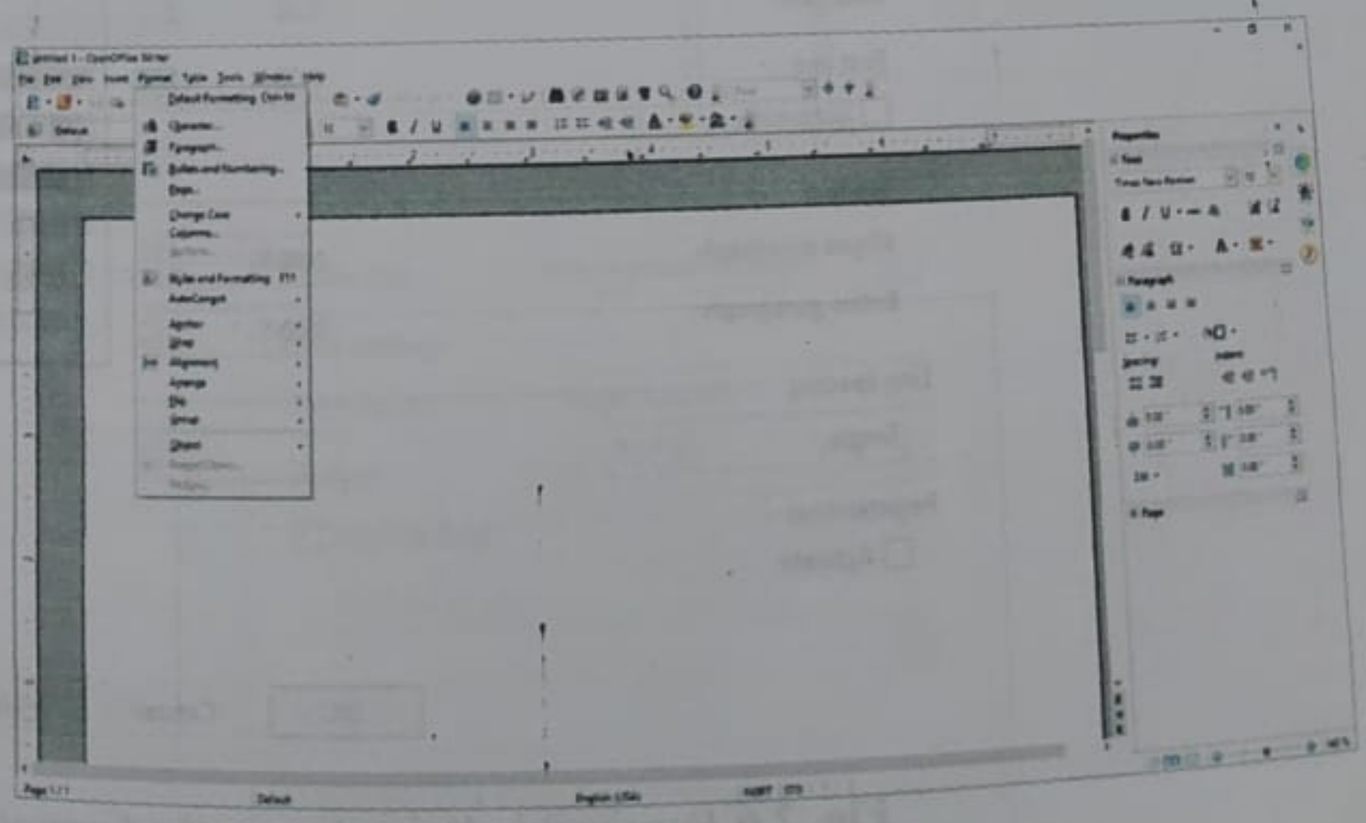


Fig. 2.4 *File menu in OpenOffice Writer*

Hyperlink and Background. Character dialog box is shown in Fig. 2.5.

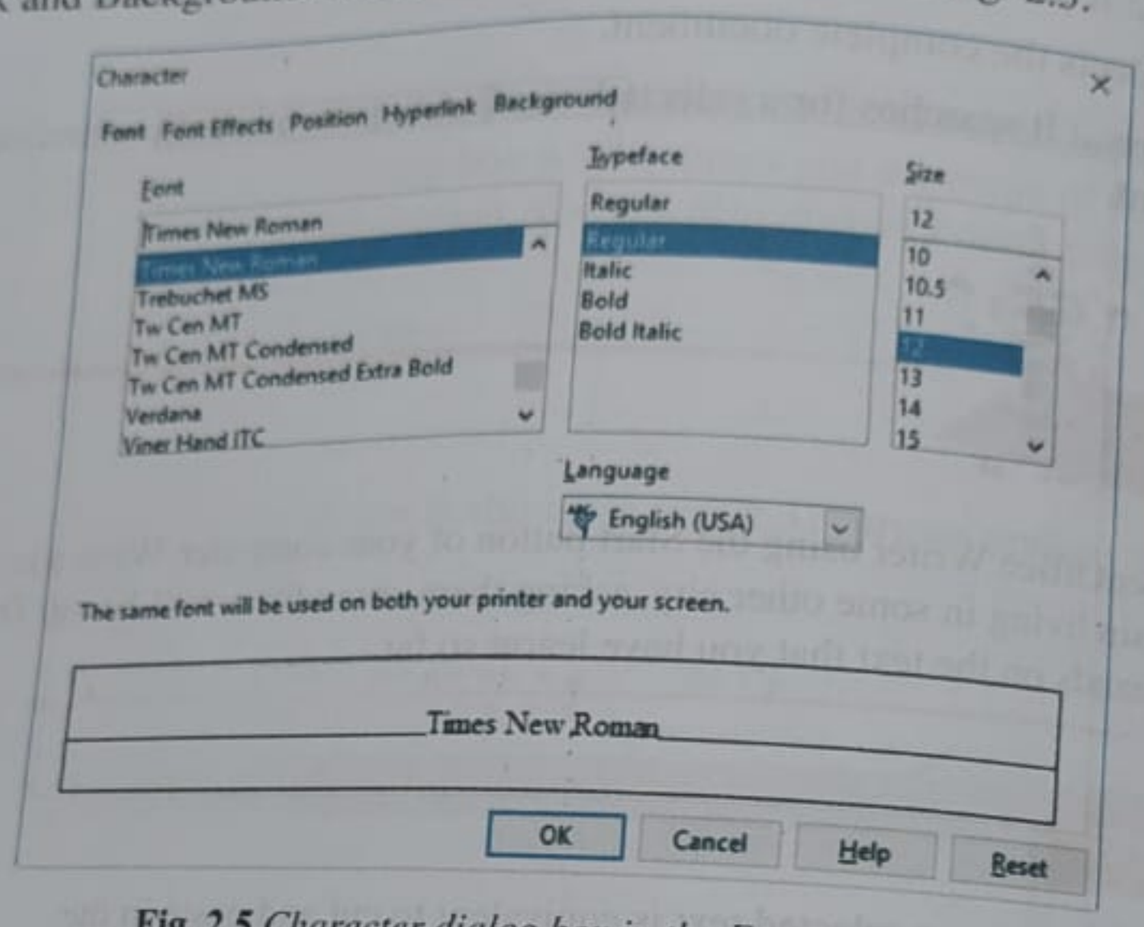


Fig. 2.5 *Character dialog box in the Format menu*

Paragraph...: The formatting related to paragraph like Indents & Spacing, Alignment, Text Flow, Outline & Numbering, Tabs, Drop Caps and Borders can be easily done as shown in Fig. 2.6.

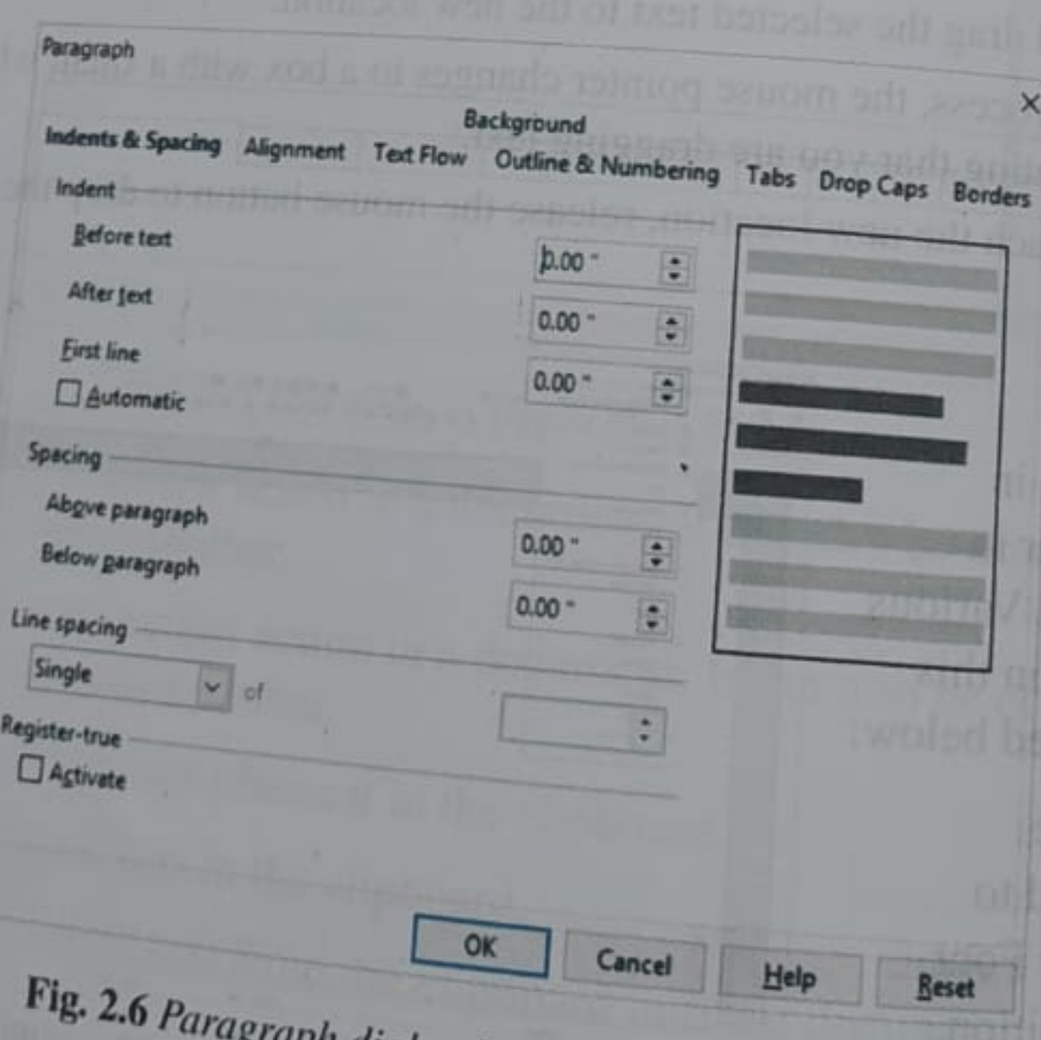


Fig. 2.6 *Paragraph dialog box in the Format menu*

Bullets and Numbering...
and numbering. OpenOffi
numbers as shown in Fig.

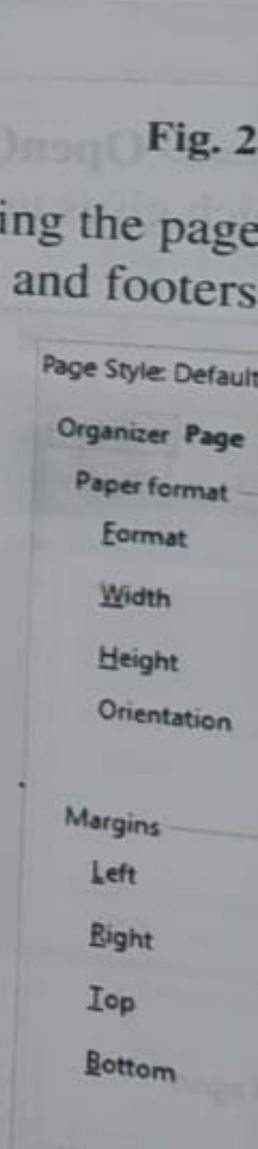


Fig. 2

Page...: Changing the page
adding headers and footers

Bullets and Numbering...: An ordered or unordered list can be created by using bullets and numbering. OpenOffice Writer has different ways of representing bullets and numbers as shown in Fig. 2.7.

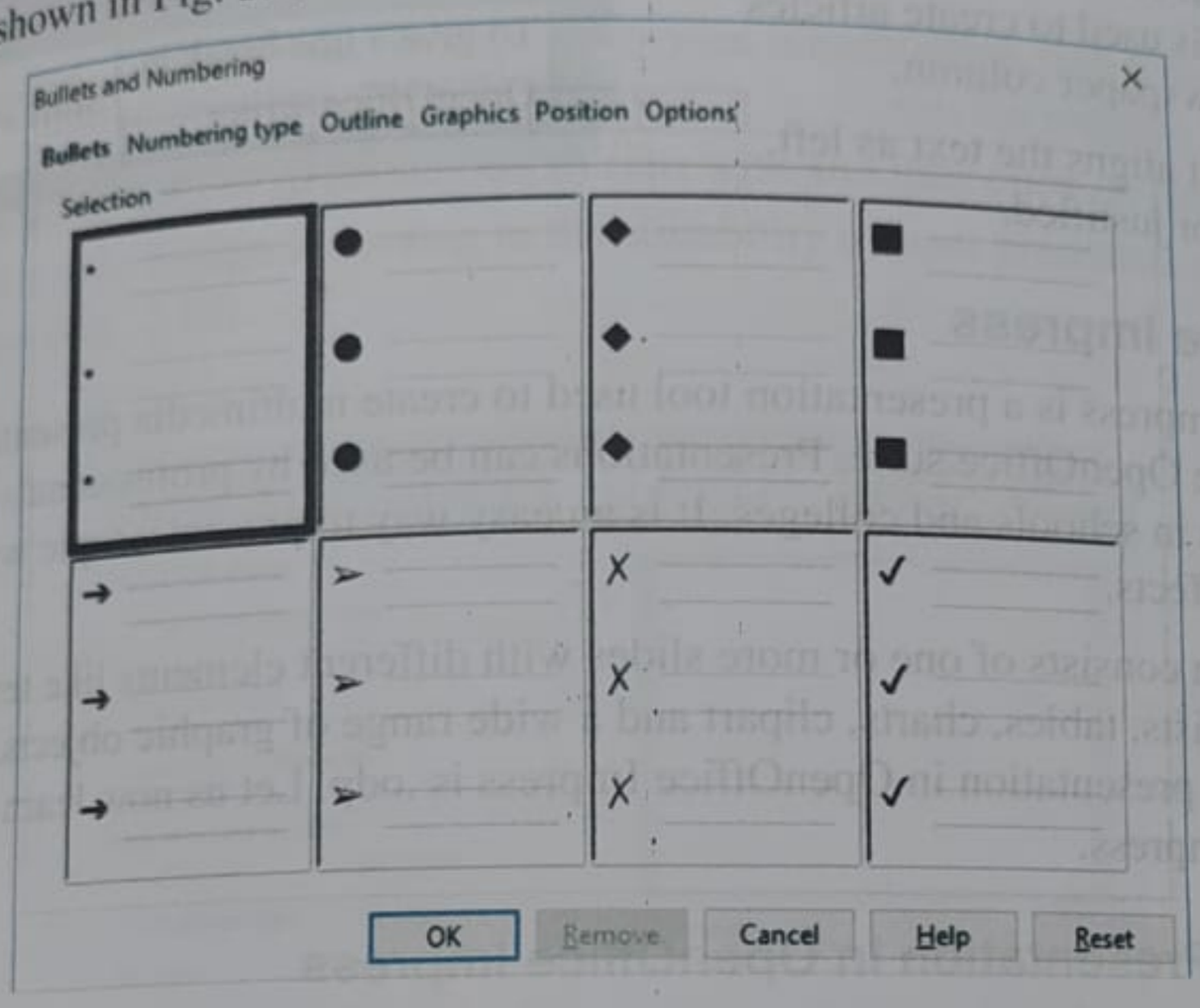


Fig. 2.7 *Bullets and Numbering in the Format menu*

Page...: Changing the page borders and background, margins, applying columns and adding headers and footers etc. can be easily done as shown in Fig. 2.8.

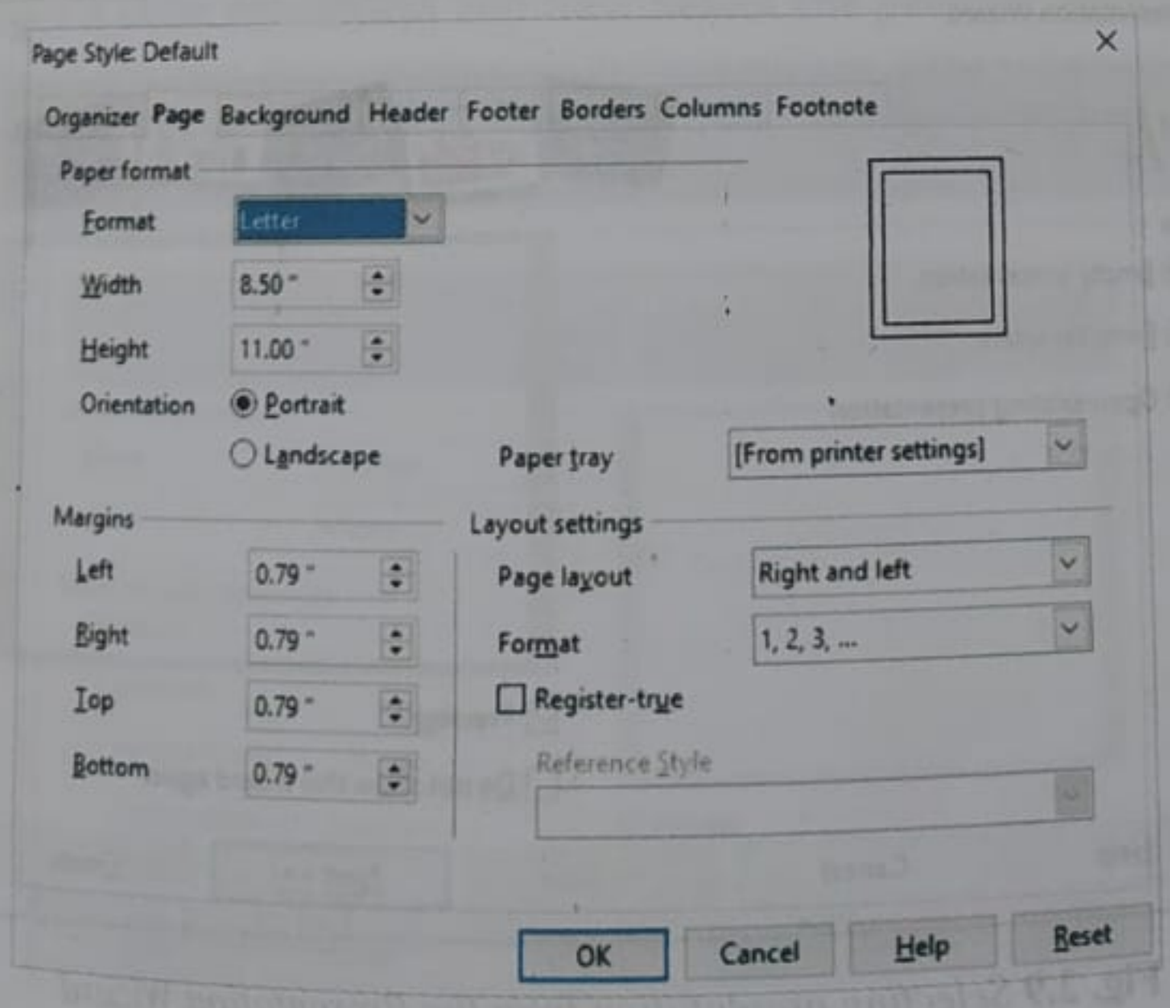


Fig. 2.8 *Page Style dialog box*

Page Preview: Previews the document before taking a printout.

Print...: Prints a document.

Printer Settings...: Changes the printer settings for a document to be printed. When clicked, it opens Printer Setup dialog box which shows you the name of the Printer and its details like Status, Type and Location. You can also change Printer properties form here.

Exit: To exit from OpenOffice Writer.

Edit Menu

The Edit menu in OpenOffice Writer is shown in Fig. 2.3. Various options available in this menu are discussed below:

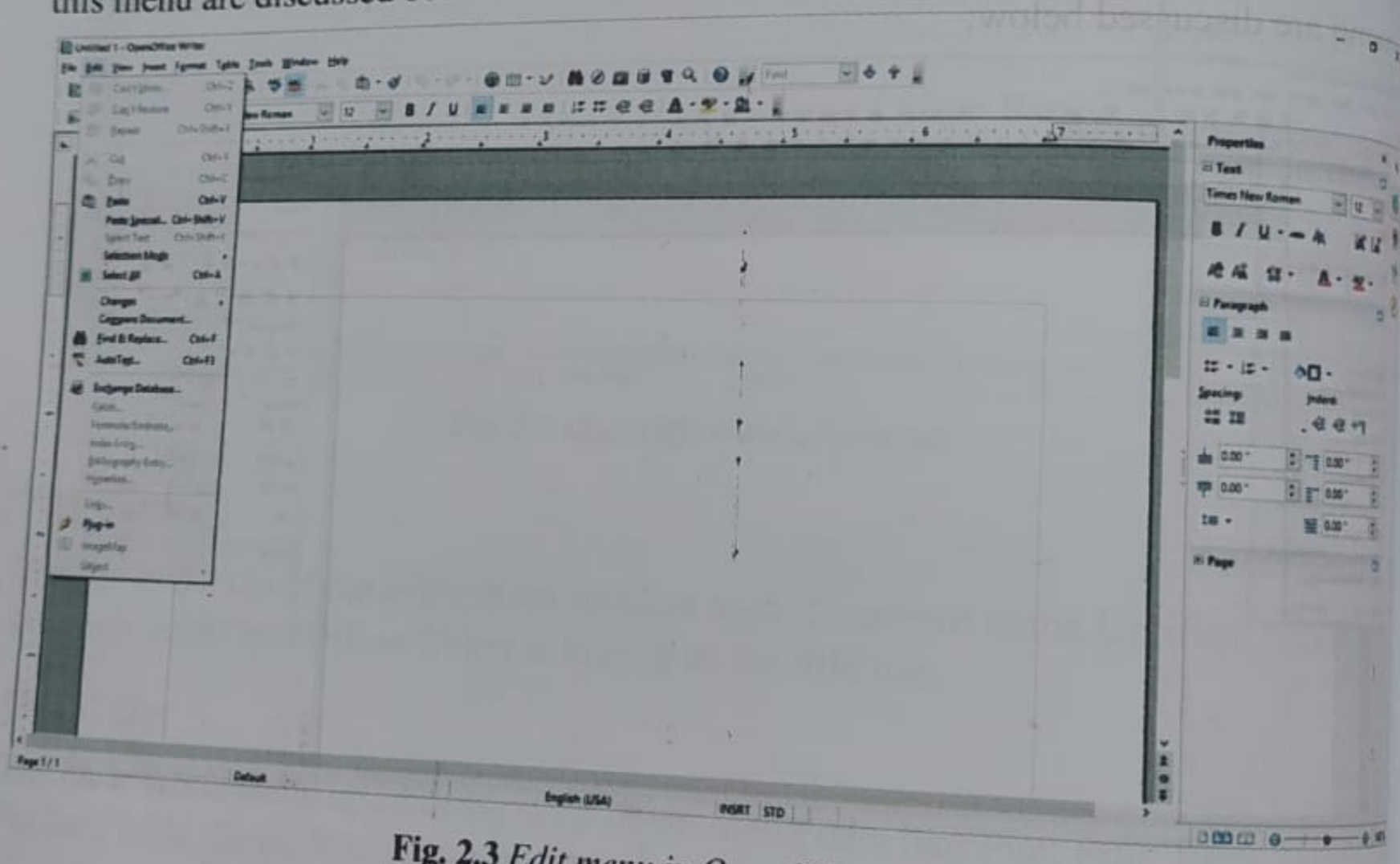


Fig. 2.3 *Edit menu in OpenOffice Writer*

Undo: It is used to undo the last action in a document. It can also be done by using the undo tool from the standard toolbar.

Restore: It is used to redo the last action in a document. It can also be done by using the redo tool from the standard toolbar.

Cut: It cuts the selected text and places it in the clipboard.

Copy: It copies the selected text in the clipboard.

Paste: It pastes the cut/copied text from the clipboard into the document.

- **Empty presentation**: It creates a blank presentation.
- **From template**: It provides some predefined templates from where we can choose and add data by ourselves.
- **Open existing presentation**: It helps us to open presentations which were created earlier.

1. Let us select **Empty presentation** to start with and then click on **Next>>**.
2. **Select a slide design** according to the suitability of your presentation. Click on **Next>>** (Fig. 2.10).

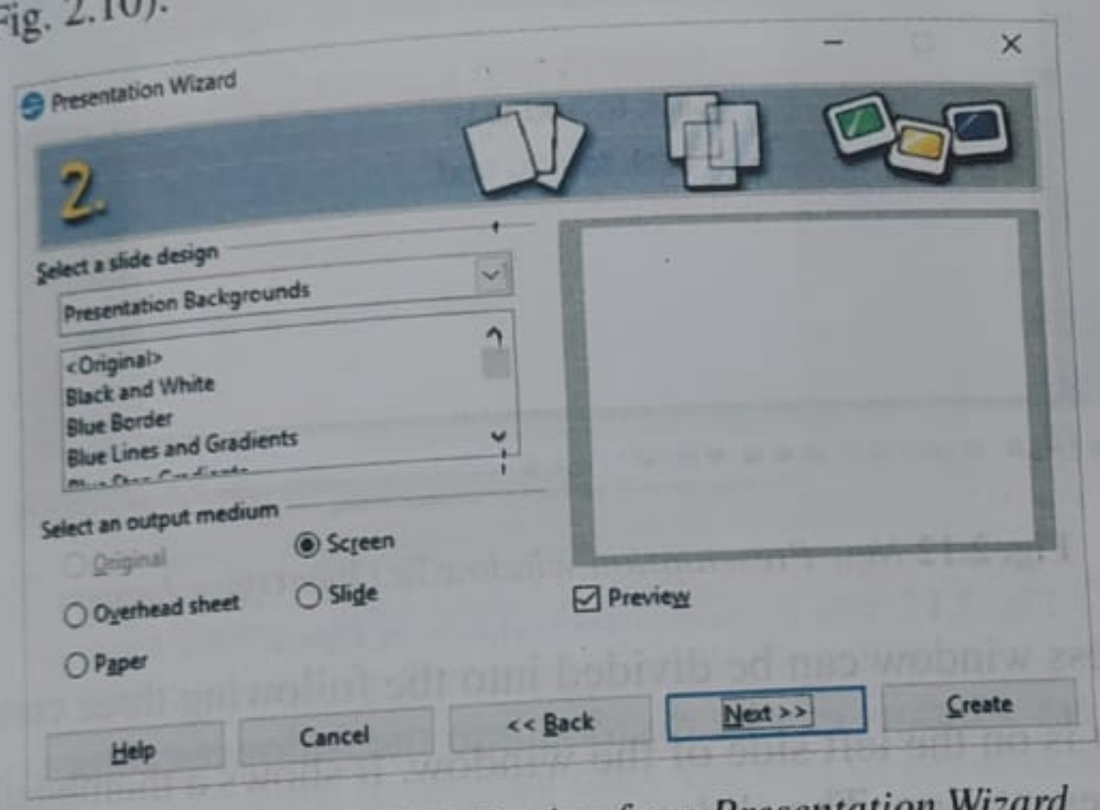


Fig. 2.10 *Selecting Slide Design from Presentation Wizard*

3. First, **Select a slide transition** and then **Select the presentation type**. Click on **Create** to create a presentation with the settings done using this 3 step presentation wizard (Fig. 2.11).

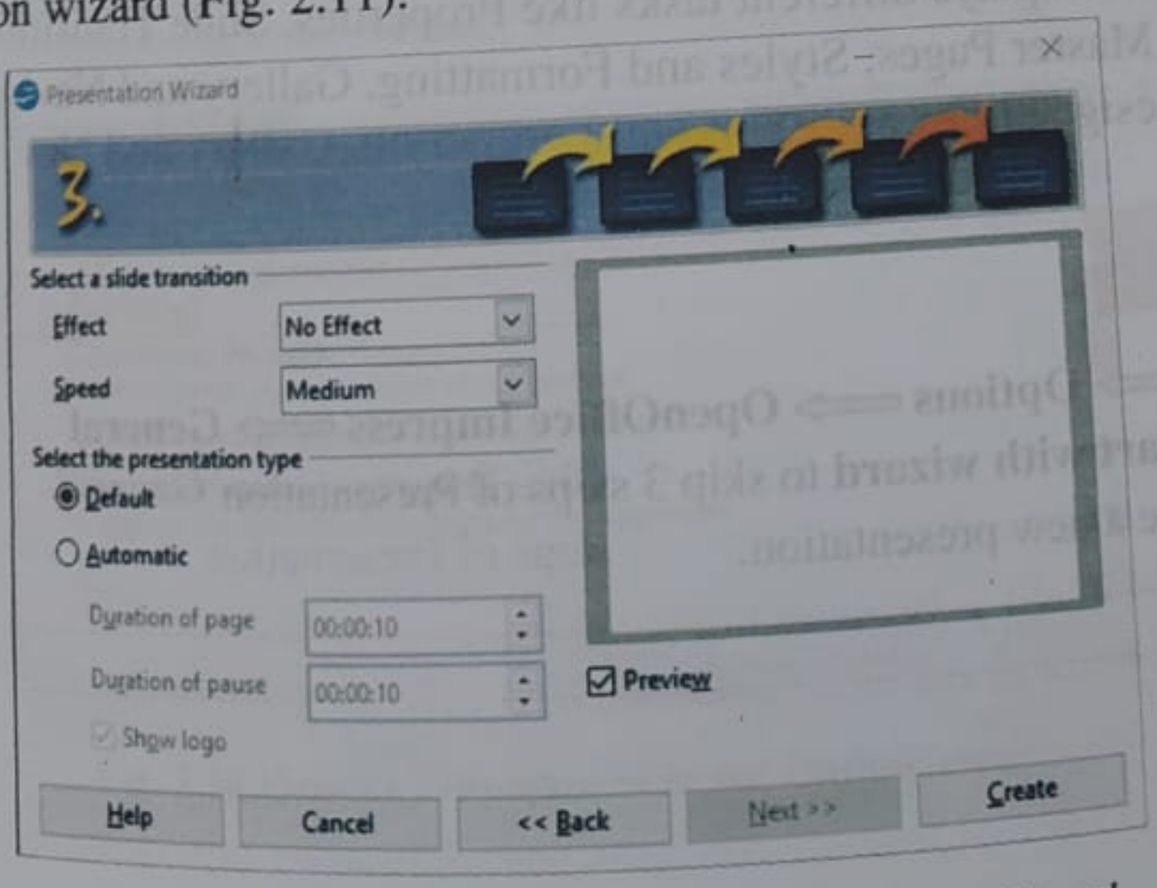


Fig. 2.11 *Selecting Slide Design from Presentation Wizard*

4. After you click on create you will get the OpenOffice Impress window as shown in Fig. 2.12.

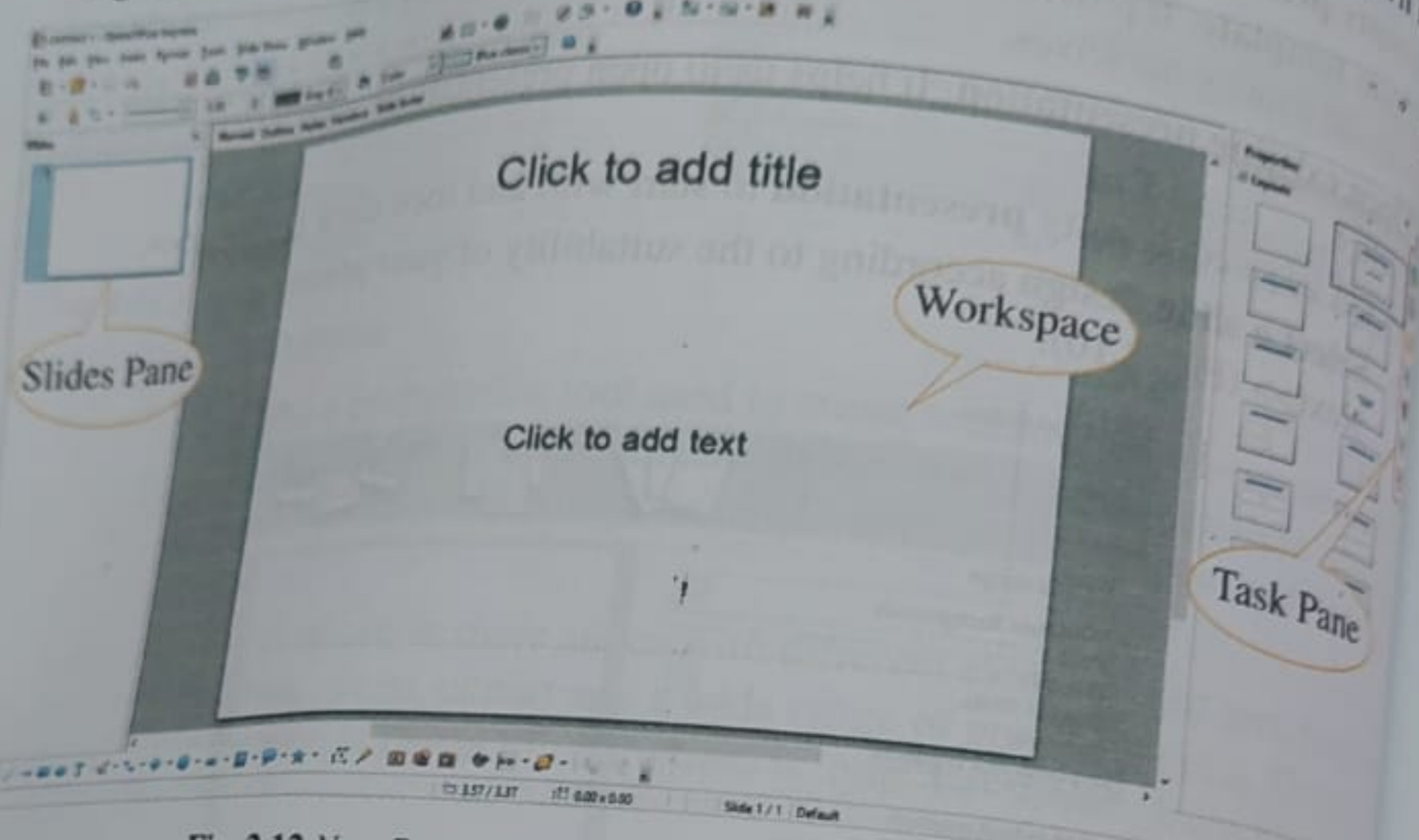


Fig. 2.12 *New Presentation window in OpenOffice Impress*

OpenOffice Impress window can be divided into the following three components:

- **Slides Pane**: It is on the left side of the window. It shows a thumbnail of all the slides in the presentation. The slides are in the order of creation. If we want we can drag and change the order of the slides.
- **Workspace**: It is the main area where we can design our slides.
- **Task Pane**: It displays different tasks like Properties, Slide Transition, Custom Animation, Master Pages, Styles and Formatting, Gallery and Navigator which can be used to design slides.

TRY THIS

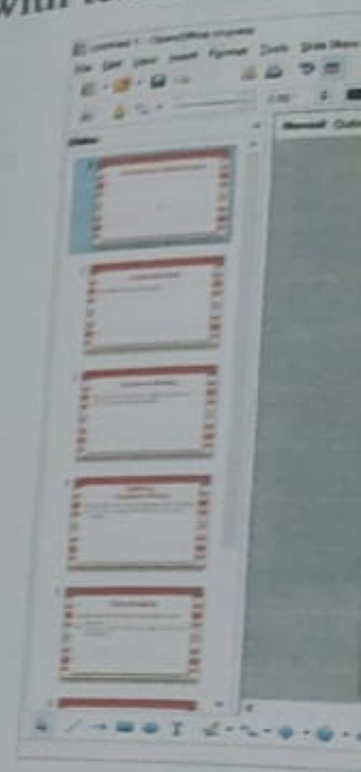
Select **Tools** ⟹ **Options** ⟹ **OpenOffice Impress** ⟹ **General** ⟹ untick **Start with wizard** to skip 3 steps of Presentation Wizard to create a new presentation.



Presentation Vie

The workspace windo
are:

- **Normal**: A slide c
with text and grap



- **Outline**: This view c
to plan the text and r

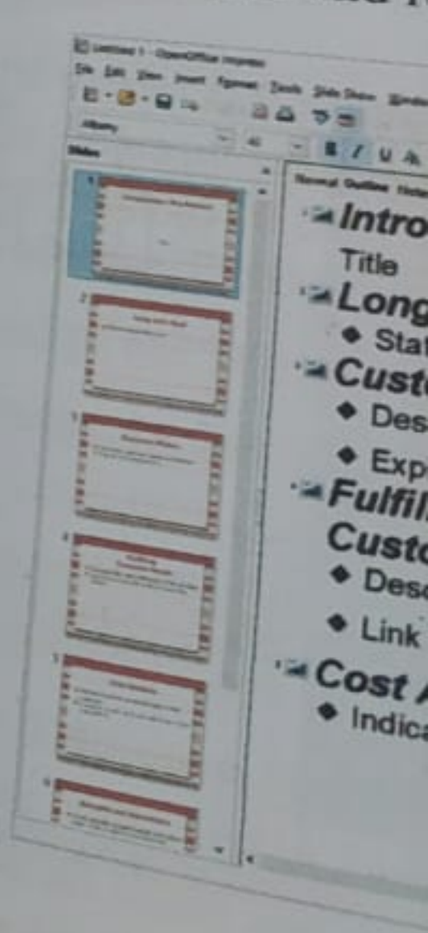


Fig.

Change Case: It converts the text into uppercase, lowercase, toggle case, title case or sentence case.

Columns...: It creates multiple columns on a page. It is used to create articles just like a newspaper column.

Alignment: It aligns the text as left, right, center or justified.

QUICK KEY

To give a line break in OpenOffice writer	Shift + Enter

OpenOffice Impress

OpenOffice Impress is a presentation tool used to create multimedia presentations. It is a part of the OpenOffice suite. Presentations can be used by professionals in offices or by students in schools and colleges. It is an easy way to present a topic with lots of multimedia effects.

A presentation consists of one or more slides with different elements like text, bulleted or numbered lists, tables, charts, clipart and a wide range of graphic objects. Default extension of a presentation in OpenOffice Impress is .odp. Let us now learn how to use OpenOffice Impress.

Creating a Presentation in OpenOffice Impress

To create a presentation, click on **Start ⟹ Openoffice.org ⟹ OpenOffice Impress**. It opens by showing the wizard given in Fig. 2.9, which gives us three options.

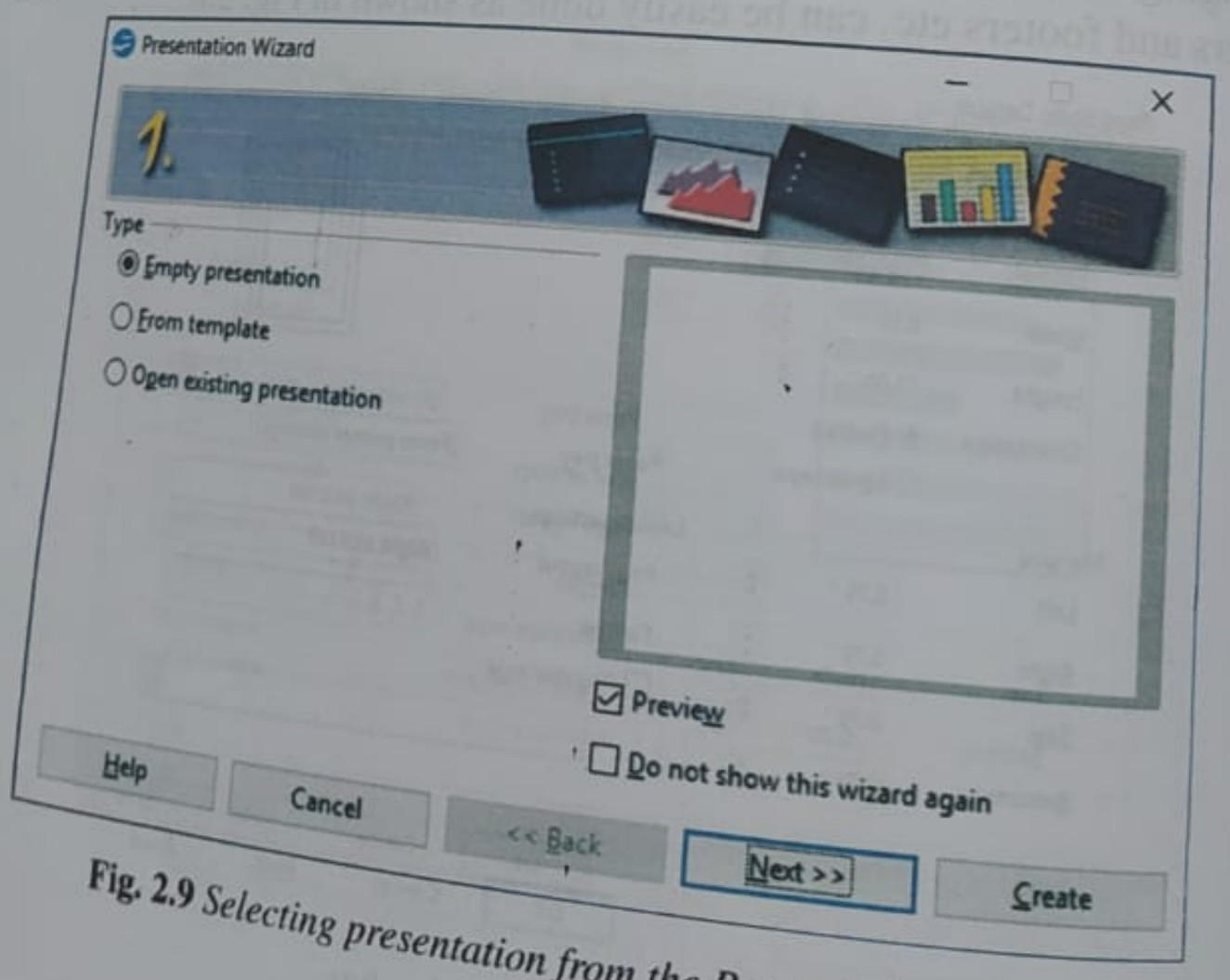


Fig. 2.9 *Selecting presentation from the Presentation Wizard*

- **Notes**: This view is used to add extra notes to the slide. It acts as reference for extra information for the audience during the slide show (Fig. 2.15).

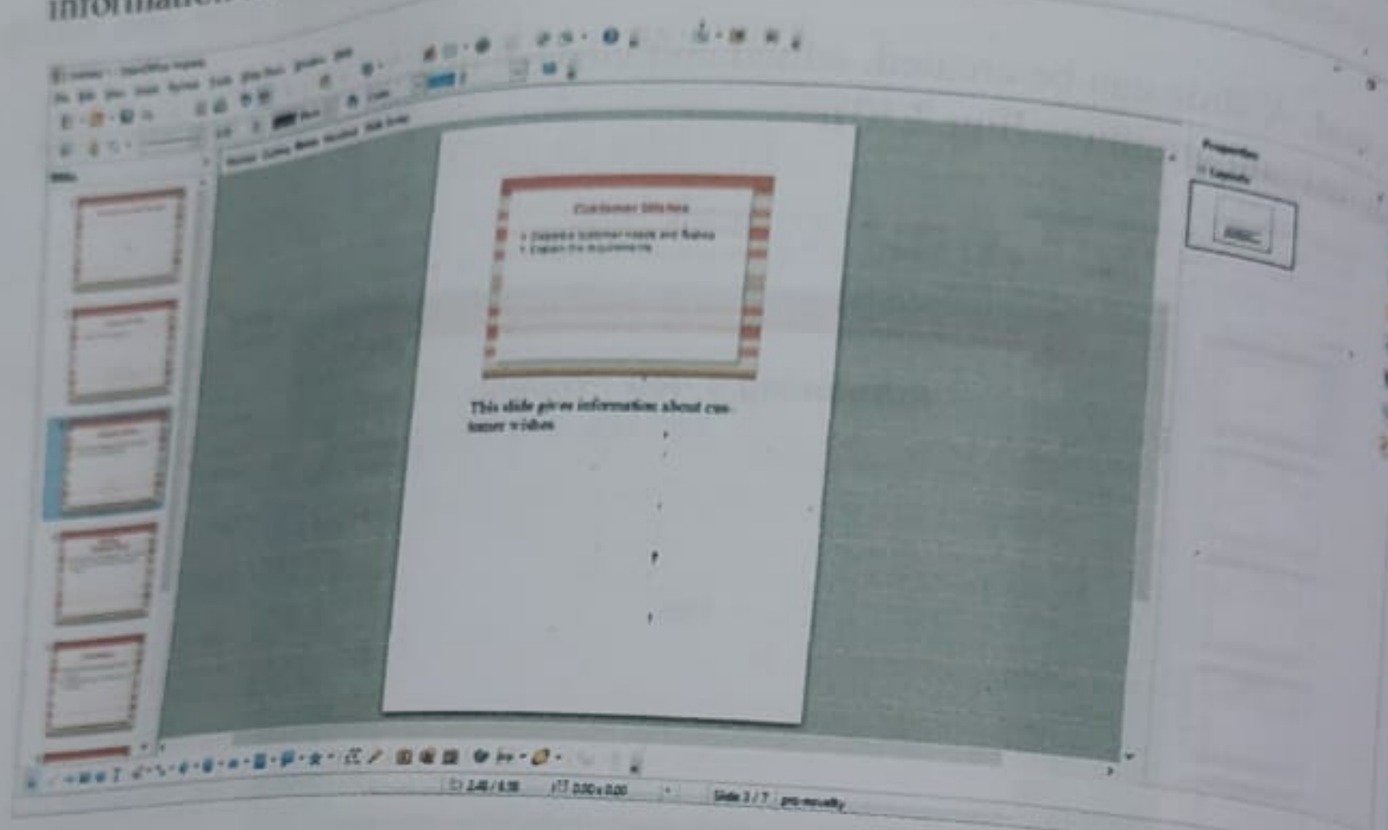


Fig. 2.15 *Viewing Presentation in the Notes view*

- **Handout**: This view creates a handout with more slides on one page. It displays the layout with single, two, three, four, six or nine slides per page (Fig. 2.16).

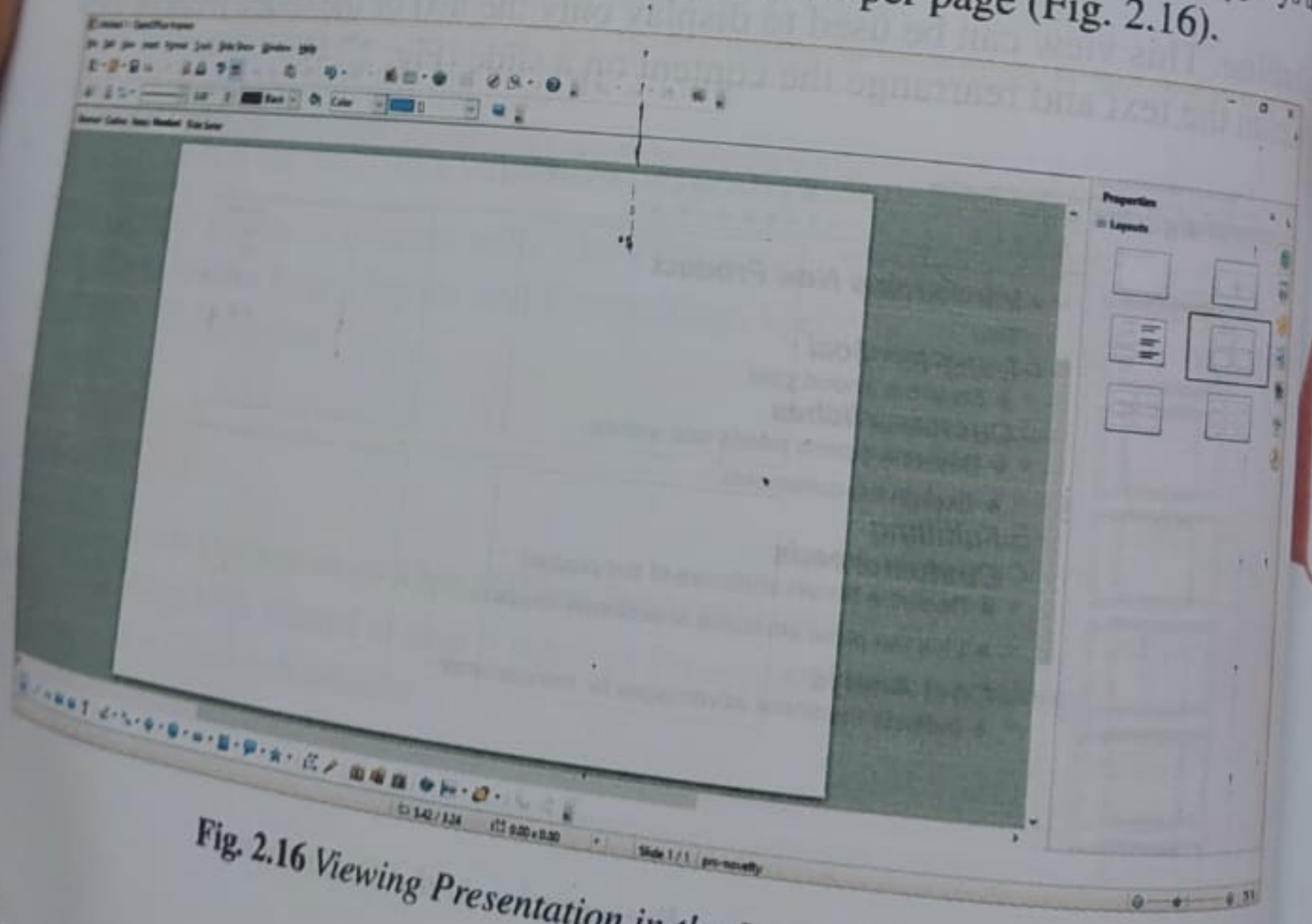


Fig. 2.16 *Viewing Presentation in the Handout view*

- ***Slide Sorter***: It displa
can rearrange the slide

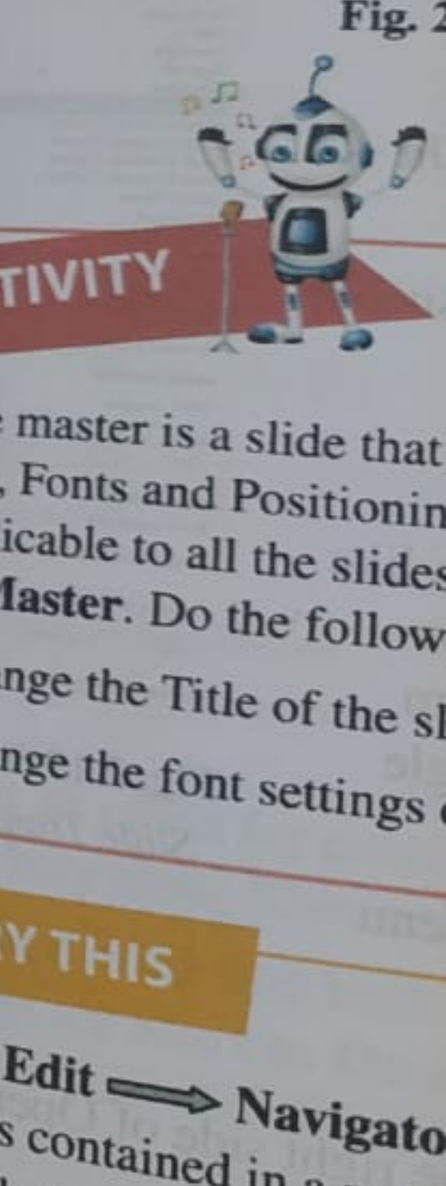


Fig. 2

ACTIVITY

A slide master is a slide that
Colour, Fonts and Positionin
be applicable to all the slides
Slide Master. Do the follow

- Change the Title of the sl
- Change the font settings

TRY THIS

Go to **Edit** ⟹ **Navigato**
objects contained in a prese
around all slides and its obj
on the standard toolbar.

- **Slide Sorter**: It displays all the slides of the presentation in a miniature form. We can rearrange the slides in this view (Fig. 2.17).

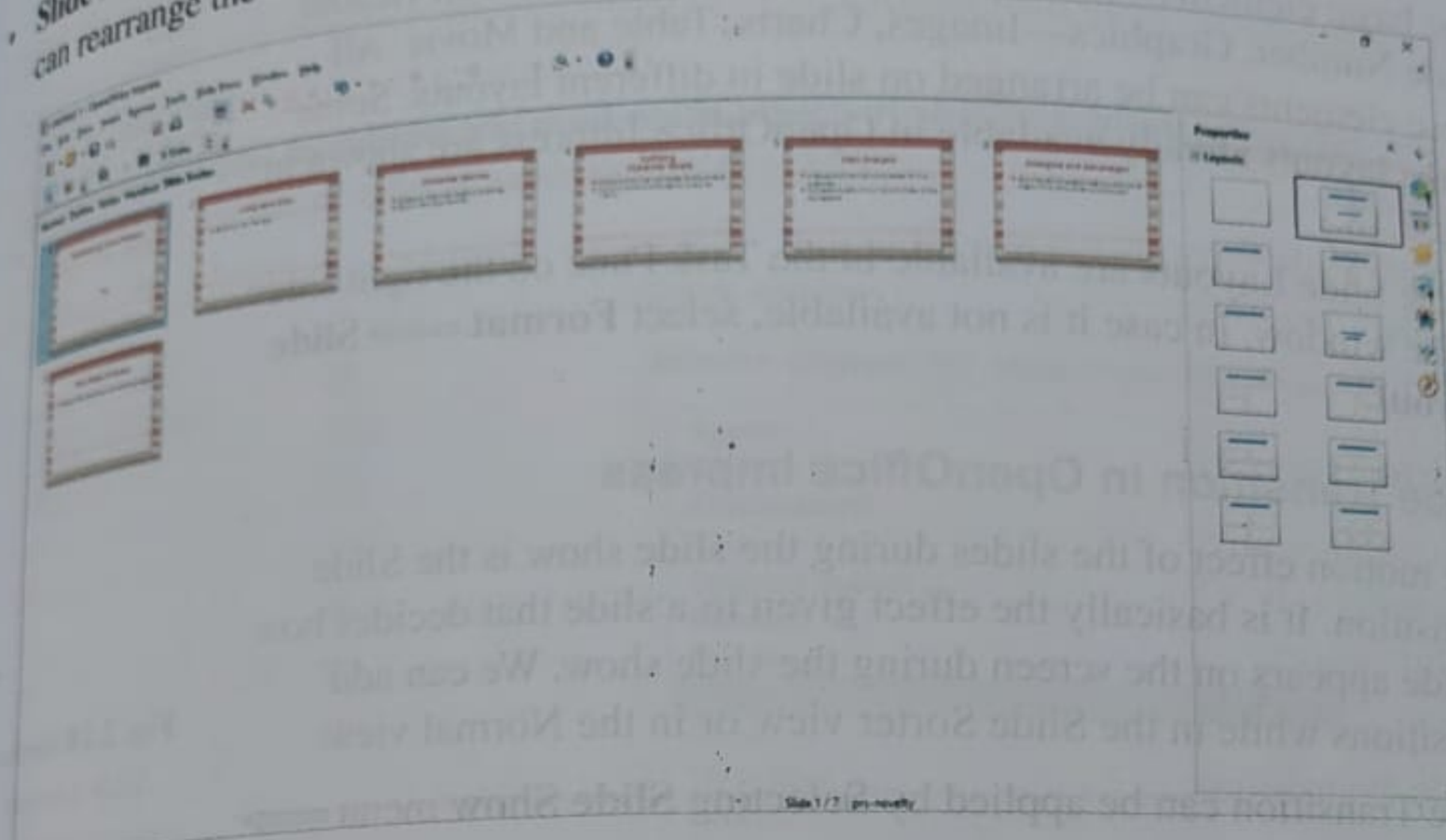


Fig. 2.17 *Viewing Presentation in the Slide Sorter view*



ACTIVITY

A slide master is a slide that controls all information about the Theme, Layout, Background, Colour, Fonts and Positioning of all slides. The settings that we do in Slide Master will be applicable to all the slides created in a presentation. Go to **View ⟹ Master ⟹ Slide Master**. Do the following changes in the Slide Master:

- Change the Title of the slides to font size = 50, bold, "Comic San MS", Colour = Purple.
- Change the font settings of the content to different bullets, colours, sizes and types.

TRY THIS



Go to **Edit ⟹ Navigator** in OpenOffice Impress to display all the objects contained in a presentation. It provides a convenient way to move around all slides and its objects. You can also use navigator button present on the standard toolbar.

Presentation Views in OpenOffice Impress

The workspace window in OpenOffice Impress provides five different view tabs, which are:

- **Normal**: A slide can be created, edited and viewed. It displays the complete slide with text and graphics (Fig. 2.13).

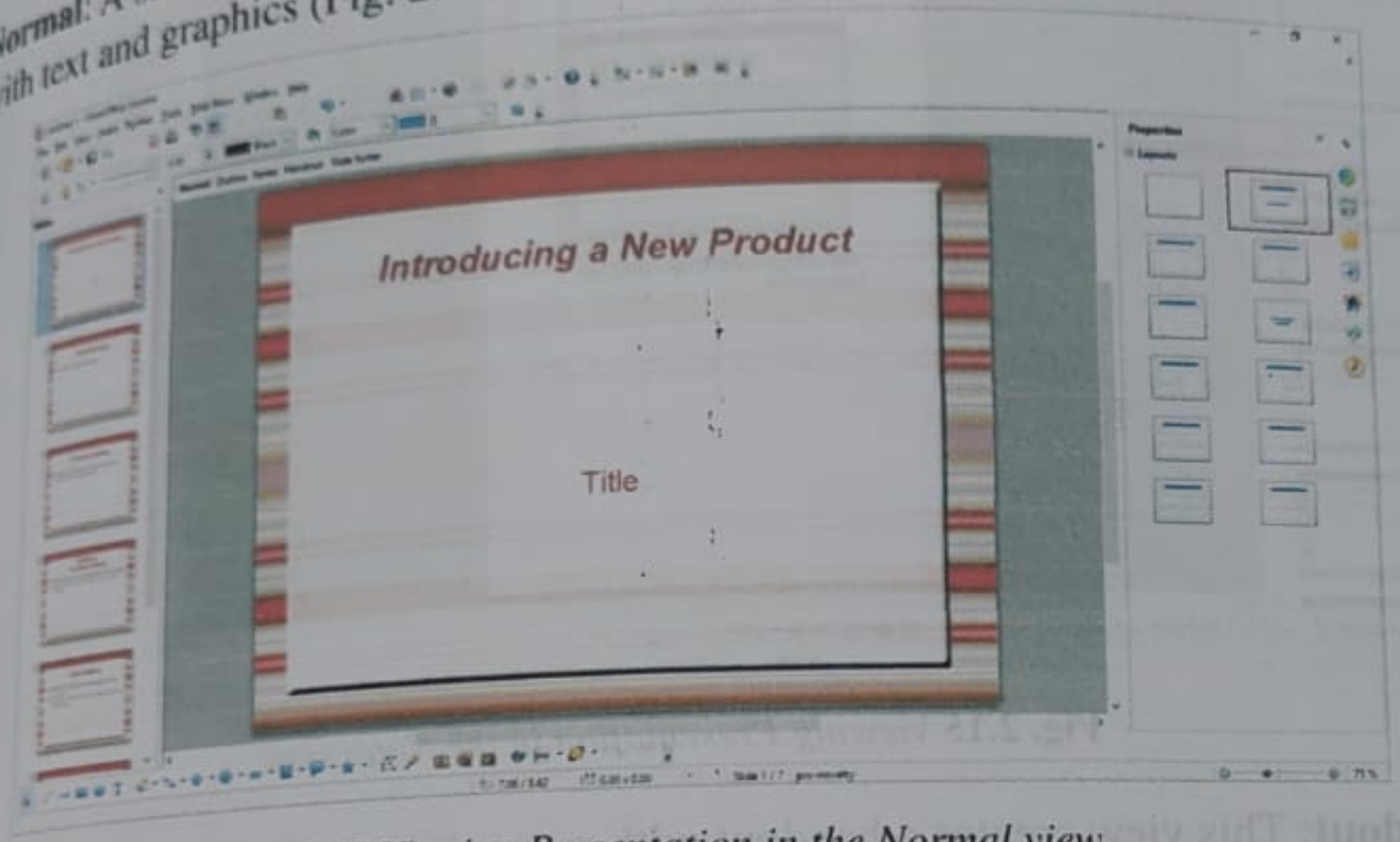


Fig. 2.13 *Viewing Presentation in the Normal view*

- **Outline**: This view can be used to display only the text of the slides. It can be used to plan the text and rearrange the content on a slide (Fig. 2.14).

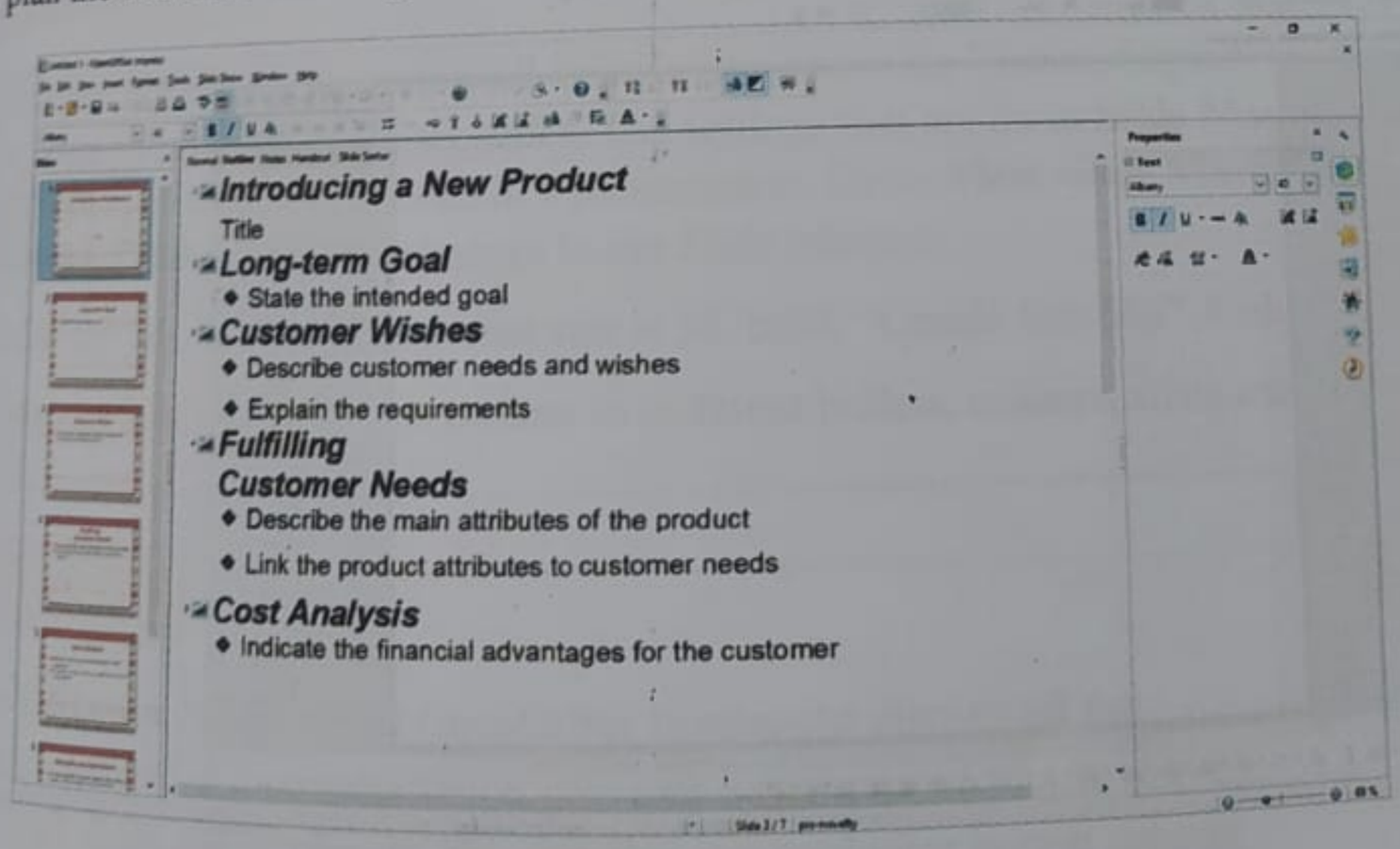


Fig. 2.14 *Viewing Presentation in the Outline view*

Steps to Apply Custom Animation

1. Select the element to which an animation is to be applied then click on the **Custom Animation** tab.
2. Click on the **Add...** button to add animation effects (Fig 2.20). It opens another window with the list of effects as shown in Fig 2.21.

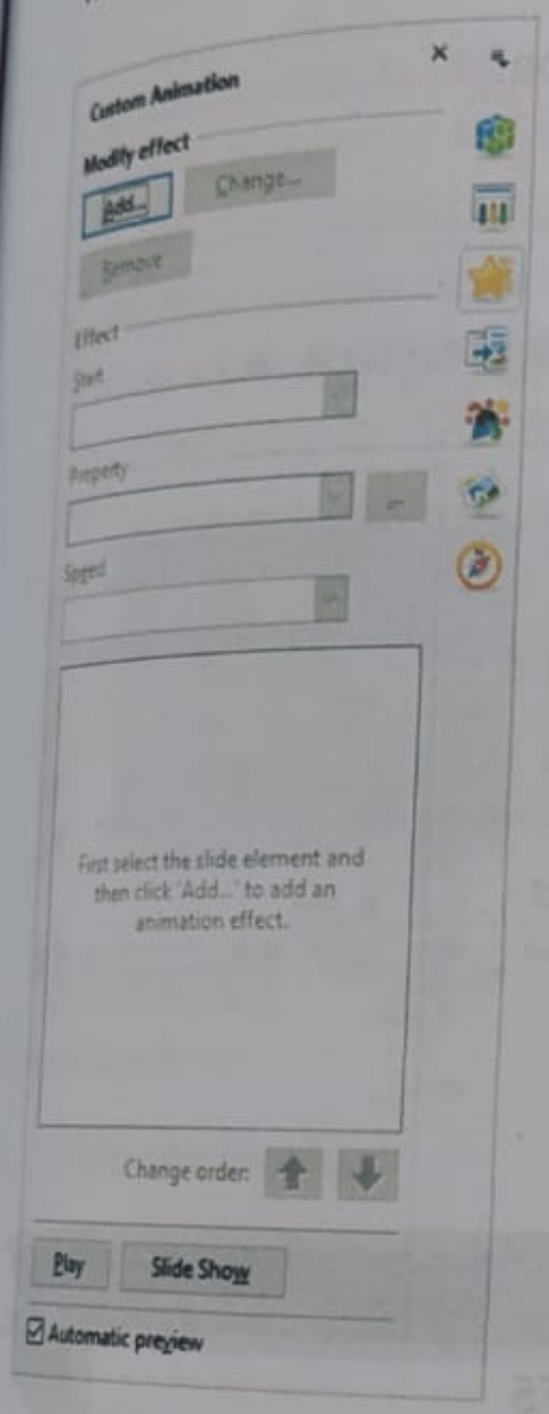


Fig. 2.20 *Custom Animation window*

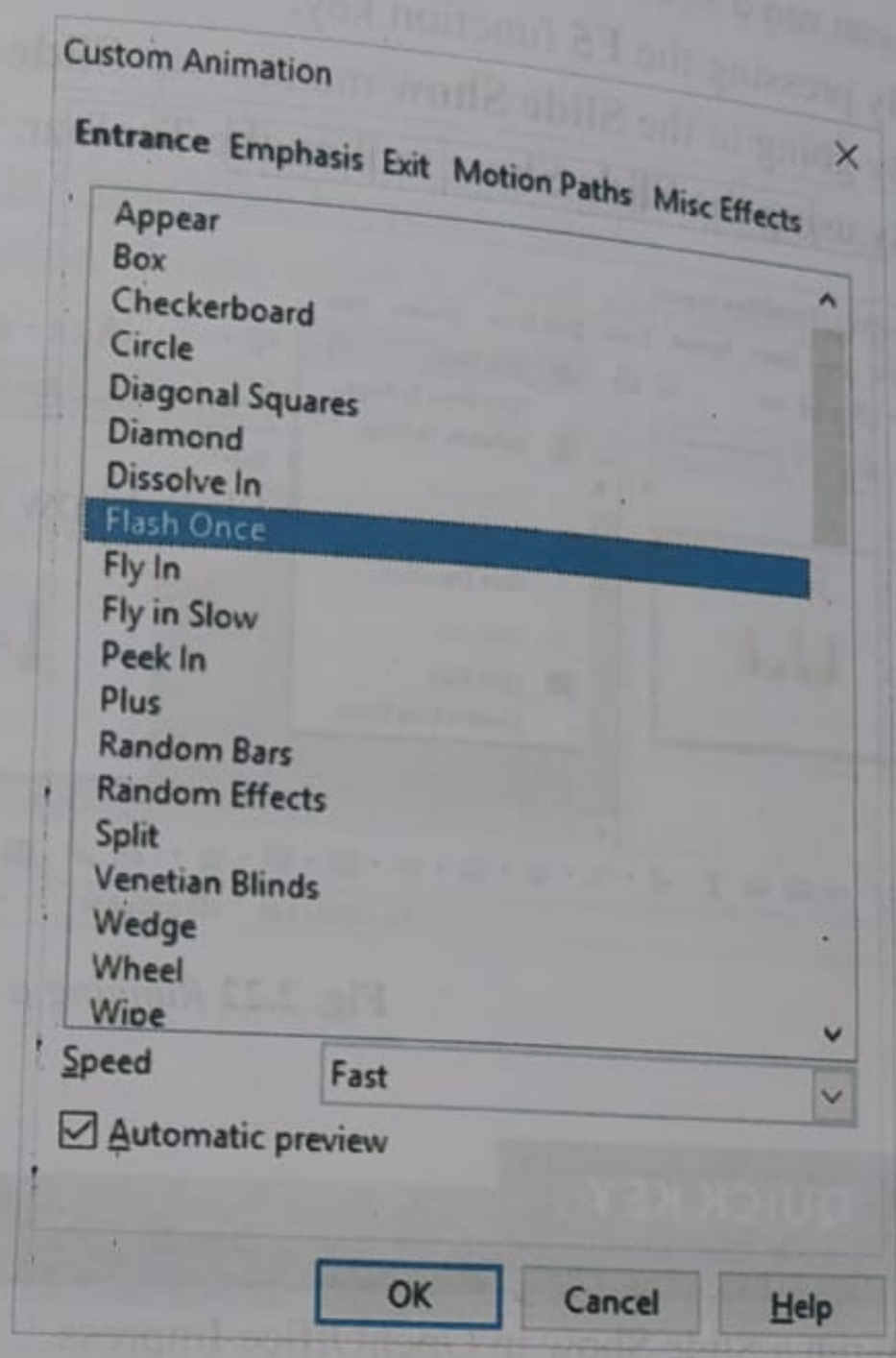


Fig. 2.21 *Selecting Custom Animation*

3. Select an effect for example, **Flash Once**. Select the **speed** of the effect as Very fast, Fast, Very slow, Slow and Medium. Click on OK.
4. This will bring you back to the Custom Animation window. Now you can select options for **Start** as **On click**, **With previous** or **After previous** depending upon whether you wish to start the animation by clicking the object or with the previous effect or after the previous effect has stopped playing.
5. Click on the **Play** button to see the animation effect applied.

Running a Presentation in OpenOffice Impress

While running a presentation, the slides appear one by one on the screen showing the transition of the slides with the animated text and graphics created earlier. This is called a **slide show** of the presentation.

You can run a slide show in three different ways.

- By pressing the **F5** function key.
- By going to the ***Slide Show*** menu ⟹ ***Slide Show*** option.
- By using the ***Slide Show*** tool on the Toolbar.

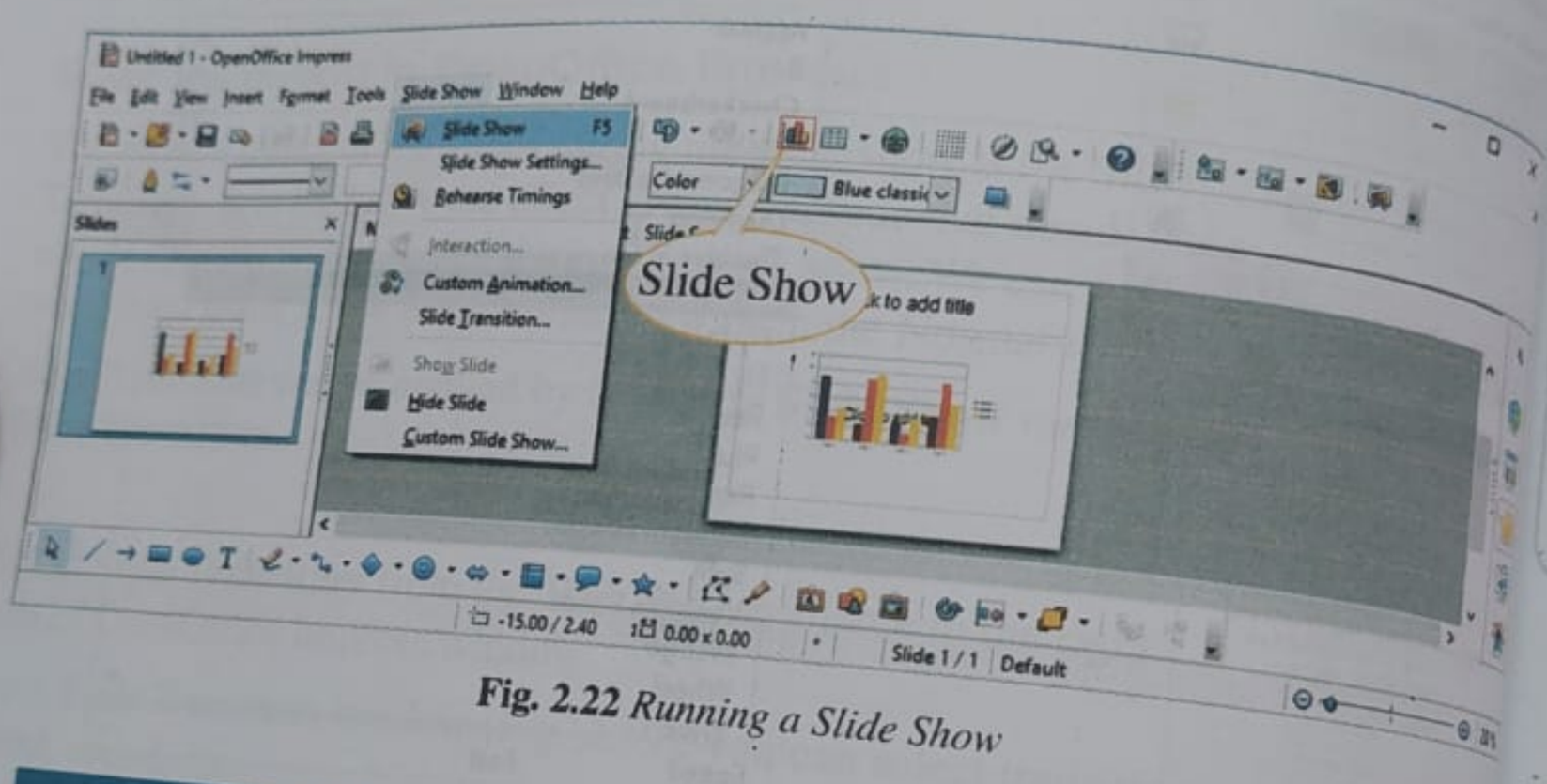


Fig. 2.22 *Running a Slide Show*

QUICK KEY

To run a Slide Show in OpenOffice Impress	**F5**

GLOSSARY

OpenOffice It is an open source software application which can be used on any operating systems like Windows, Linux and MacOS.

OpenOffice Impress It is a presentation tool used to create multimedia presentations.

OpenOffice Writer It is a word processor used to create, edit, format and print a document

YOU ARE HERE 2

1. The
2. File new
3. Edit & Re
4. The Format menu Bullets & Numberi
5. The default extensi
6. OpenOffice Impres creating a new pres
7. The window of Ope Workspace and Task
8. The workspace in O Normal, Outline, No

EXERCISE

A. Fill in the blanks.

1. Cut and Paste ar
2. To create and ed
3. The .....
4. The .....
5. The .....

B. State true or false.

1. The file extension
2. Slides can be easil
3. F6 is used to run a

Slide Layouts in OpenOffice Impress

The basic elements of the slides are Title, Content, Footer, Header, Slide Number, Graphics—Images, Charts, Table and Movie. All these elements can be arranged on slide in different layouts. Some of the layouts already available in OpenOffice Impress are shown in Fig. 2.18.

These Slide Layouts are available in the Task Pane on the right side of the window. In case it is not available, select **Format** ⟹ **Slide Layout**.

Slide Transition in OpenOffice Impress

The motion effect of the slides during the slide show is the Slide Transition. It is basically the effect given to a slide that decides how a slide appears on the screen during the slide show. We can add transitions while in the Slide Sorter view or in the Normal view.

Slide Transition can be applied by Selecting **Slide Show** menu ⟹ **Slide Transition...**

OR

Select **Slide Transition** from the Task Pane available in the right side of OpenOffice Impress window.

From Slide Transition window (Fig. 2.19), you can select transition, speed, sound, etc.

Custom Animation in OpenOffice Impress

It is a set of animated effects that can be applied on the objects like title, content and graphics. Animations can make a presentation more lively and memorable. Animation effects are applied from Normal View so that we can select individual objects on a single slide.

Custom Animation can be applied by selecting **Slide Show** menu ⟹ **Custom Animation...**

OR

Select **Custom Animation** from the Task Pane available in the right side of OpenOffice Impress window.



Fig. 2.18 *Various Slide Layouts*

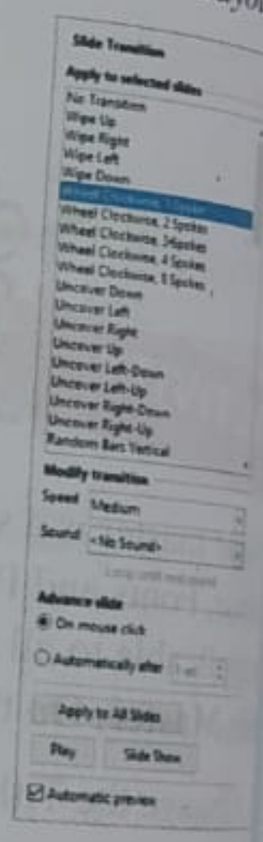


Fig. 2.19 *Selecting Slide Transitions*

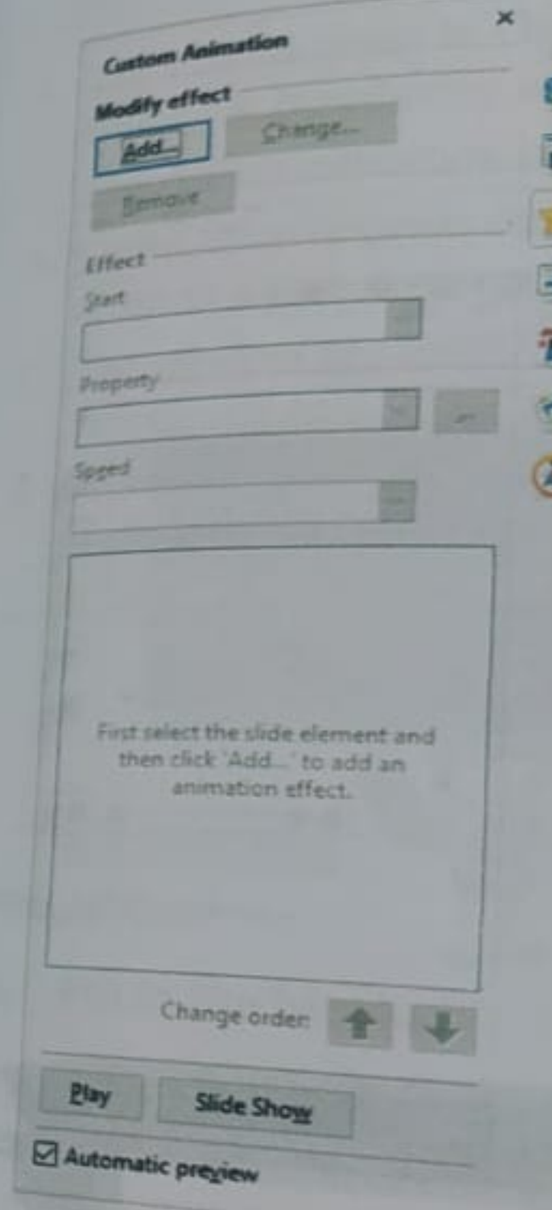


Fig. 2.20 *Custom Animation window*

Steps to Apply Custo

1. Select the element to w
   Animation tab.
2. Click on the **Add...** but
   window with the list of

3. Select an effect for examp
   Fast, Very slow, Slow and
4. This will bring you back
   options for **Start** as **On c**
   whether you wish to start
   effect or after the previou
5. Click on the **Play** button t

4. We cannot add new slides after a presentation is saved.
5. Justified alignment arranges the document on both left and right margins.
6. The Page Preview option can be used to print a document.

C. Match the following.

1. Slide Transition	a. Runs a presentation
2. F5	b. Changes font size, type, colour
3. Slide Show	c. Displays only text
4. Outline View	d. Custom animation
5. Format ⟹ Character	e. Effects on slides

D. Answer the following questions.

1. Why do we use OpenOffice Writer?
2. Explain any two options present in the Format menu of OpenOffice Writer.
3. Describe three different ways to run a presentation.
4. What is Slide Transition? Name any two effects.



LAB WORK

A. Create an OpenOffice document on the topic 'importance of healthy food'. Take the help of the Internet to get some information and pictures.

B. Create a presentation on 'The Place I visited in My Summer Holidays'. Take the help of the Internet to get some information and pictures.



PR

Work in groups of four.

Researcher: The one wh
school for a period of 15

Creative writer: One wh
OpenOffice Writer.

Designer: One who desig
Impress.

Animator: One who deci

All the groups need to cor
group presentation in fron



PROJECT WORK

Work in groups of four. Each student in the group needs to take up one of the below roles.

Researcher: The one who gathers news about various events, competitions etc. in the school for a period of 15 days. Use OpenOffice Writer to store the research work.

Creative writer: One who puts the text together in a concise and precise way using OpenOffice Writer.

Designer: One who designs the layout of the fortnightly newsletter in OpenOffice Impress.

Animator: One who decides the text and graphics effects in OpenOffice Impress.

All the groups need to come up with a fortnightly newsletter for the School and make a group presentation in front of the class using OpenOffice Applications.

owing the
his is called

YOU ARE HERE

2

1. The extension of the file created in OpenOffice Writer is .odt.
2. File menu in OpenOffice Writer consists of various options for creating a new file and saving and printing it.
3. Edit menu in OpenOffice Writer includes options to Cut, Copy, Paste, Find & Replace etc.
4. The Format menu in OpenOffice Writer provides options like Character..., Paragraph..., Bullets & Numbering and Change Case.
5. The default extension of a presentation in OpenOffice Impress is .odp.
6. OpenOffice Impress starts with Presentation Wizard, which is a three step procedure for creating a new presentation.
7. The window of OpenOffice Impress is divided into three parts, which are Slides Pane, Workspace and Task Pane.
8. The workspace in OpenOffice Impress provides five different view tabs, which are Normal, Outline, Notes, Handout and Slide Sorter.



EXERCISE

A. Fill in the blanks.

1. Cut and Paste are present in the ..... menu.
2. To create and edit a document we use OpenOffice ..... .
3. The ..... view shows miniature form of all slides in a presentation.
4. The ..... option in File menu will save the file for the first time.
5. The ..... is used to give animation to the objects on the slides.

B. State true or false.

1. The file extension of OpenOffice Impress presentation is .ods.
2. Slides can be easily rearranged and deleted in the Slide Sorter View.
3. F6 is used to run a presentation.

can have multiple sheets showing Sheet1, Sheet2 and Sheet3 as default. Each sheet is divided into a grid of rows (horizontal) and columns (vertical).

A row is labelled as numbers starting from 1, 2, 3, … with the last row as 1048576.

A column is labelled as letters starting from A, B, C, … with the last column as AMJ.

A cell is formed by an intersection of a row and a column and is referred to as A1, B6, G32, etc.

A collection of the cells together will form a **cell range**, for example, A1:A5, B1:D7,

QUICK KEY

To go to first cell of the sheet i.e. A1	**Ctrl + Home**
To reach the last cell of the sheet i.e. AMJ1048576	**Ctrl + End**

Entering Data in a Cell in OpenOffice Calc

You can fill a workbook by entering data in the cells. Data in a cell can be in the form of text, numbers or combination of both. The default alignment for text is left while for numbers, it is right.

In order to enter data in a cell, follow the steps given below.

1. Select the cell where you want to enter the values.
2. Type the value that you want in a cell.
3. Press the Enter key or click on another cell to move to the next cell.

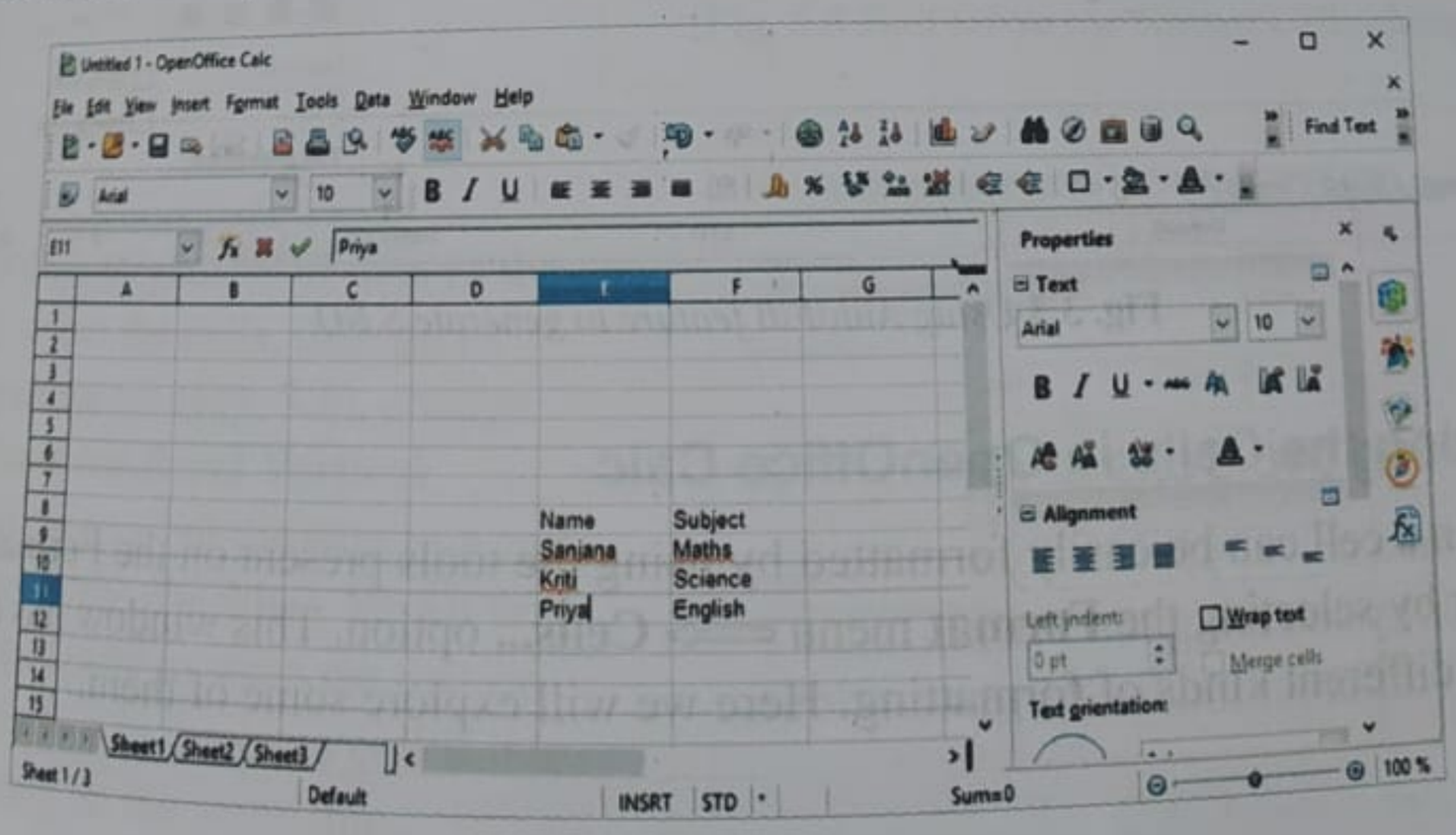


Fig. 3.2 *Entering data in cells of OpenOffice Calc window*

The cells can be formatted by changing their font name, size, colour, etc. either by using the **Formatting** toolbar or by selecting **Format** menu ⟹ **Cell...** option.

Using AutoFill Feature in OpenOffice Calc

In a spreadsheet, you can generate a series of number or text values by using the AutoFill handle. This process of automatically generating the series is called the AutoFill feature of a spreadsheet.

Follow these steps to generate the S.NO (Serial Number).

1. Write 1 in cell C5 and 2 in cell C6 and select them.
2. Click on the fill handle present at the bottom-right corner of the selected cells. Drag the mouse down till the point you want to generate the series. In the above case, drag till cell C9.
3. The series will be generated automatically as shown in Fig. 3.3.

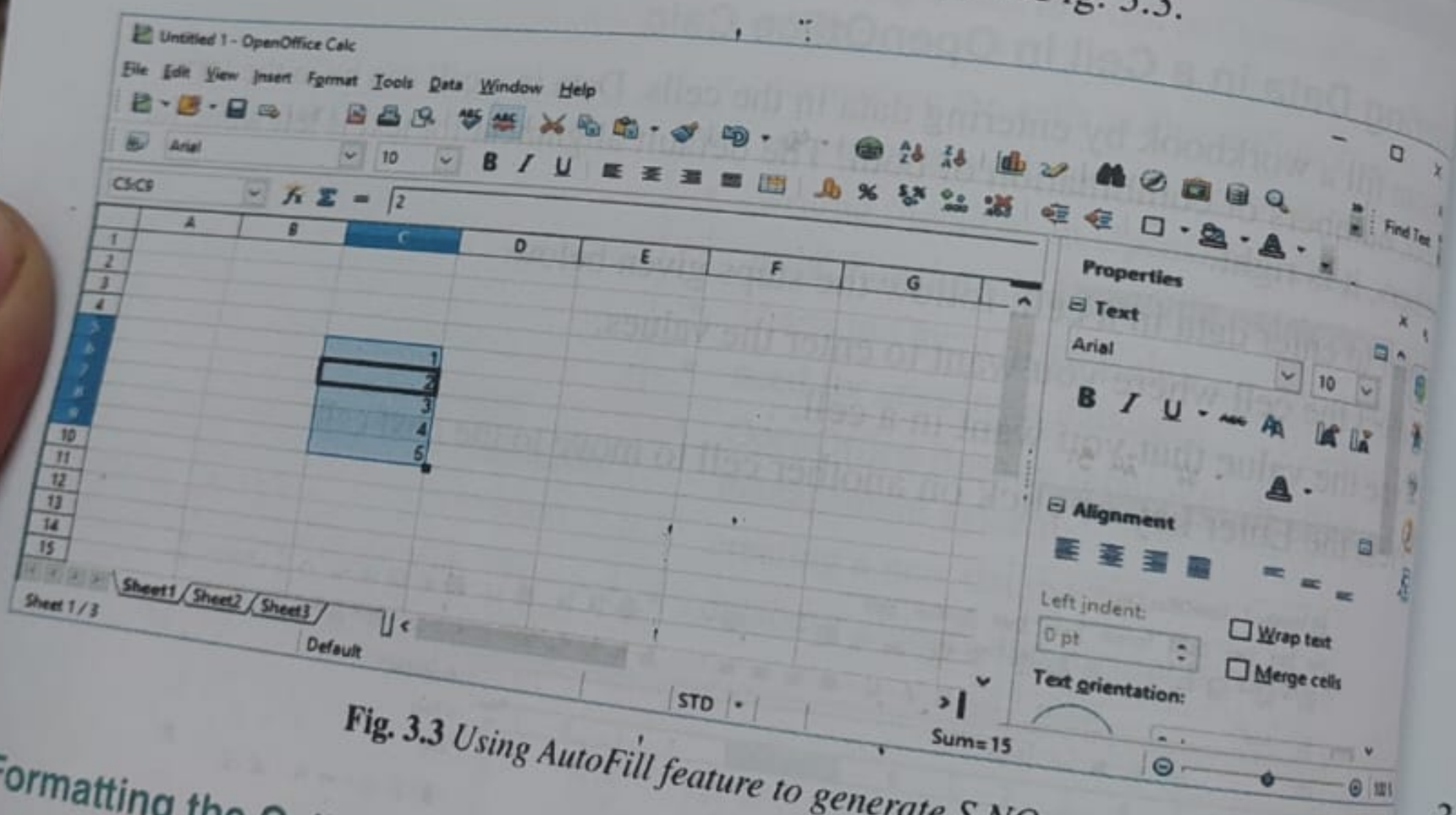


Fig. 3.3 *Using AutoFill feature to generate S.NO*

Formatting the Cells in OpenOffice Calc

The data in a cell can be easily formatted by using the tools present on the Formatting Toolbar or by selecting the **Format** menu ⟹ **Cells...** option. This window has many tabs to do different kinds of formatting. Here we will explore some of them.

1. To change the nu
by putting currenc
amount, select the
to format, then go
tab in the Cells...
Currency in **Cate**
Format code to In
Decimal Places as

2. To change the font
of the text, select the
to format, then go to
the Cells... option; s
Arial, **Typeface** as R
as 14 (Fig. 3.5).

3. To change alignment
cells, select the cells, t
Alignment tab (Fig. 3.
the **Horizontal** and **Ve**
alignment to align the
and select the checkbo
automatically to wrap
the next line in the sam
not fitting on the same

4. To change cell background, select the cells for which you want to change background, then go to the **Background** tab (Fig. 3.7), select the colour of your choice and click on the ***OK*** *button.*

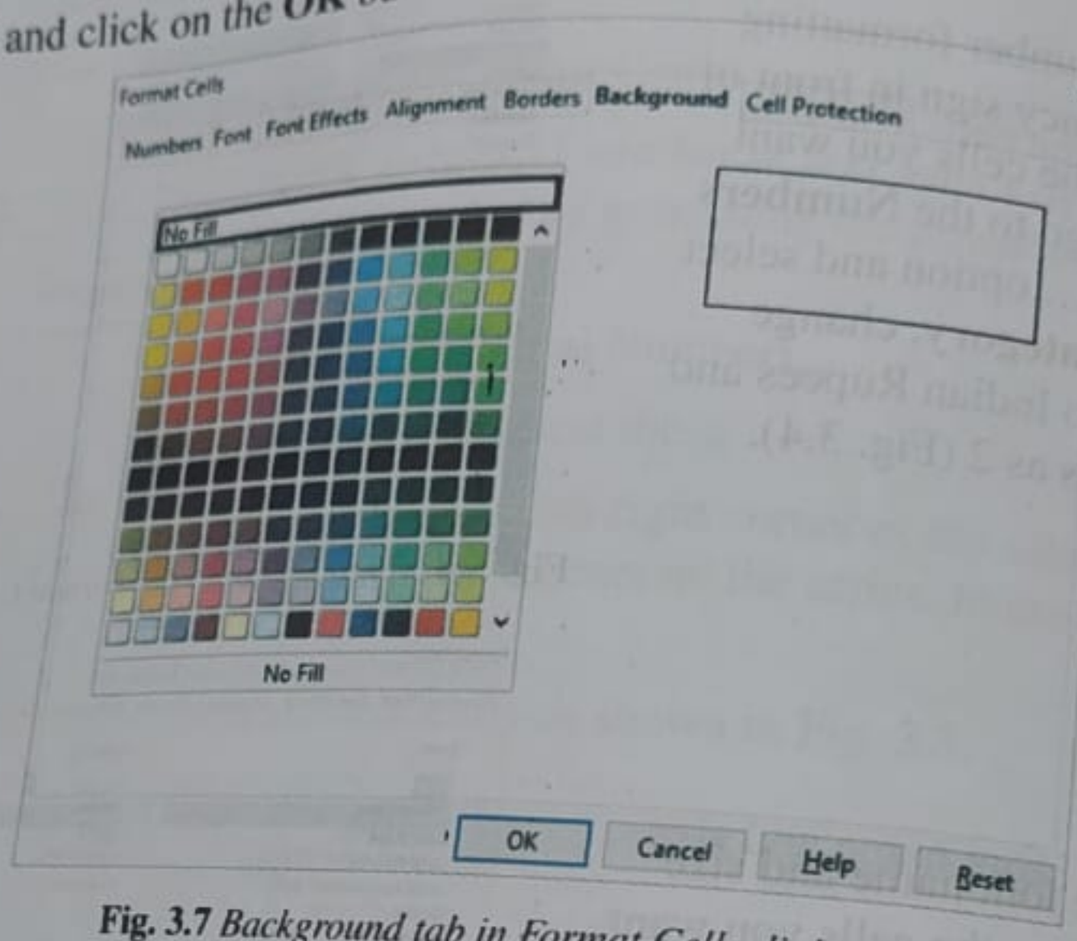


Fig. 3.7 *Background tab in Format Cells dialog box*

Renaming a Worksheet in OpenOffice Calc

The sheets are named as Sheet1, Sheet2, Sheet3, ... in OpenOffice Calc. If you want to change the name, follow the steps given below:

1. Right-click on the Sheet1 tab.
2. Select the **Rename Sheet...** option to display the given window (Fig. 3.8).

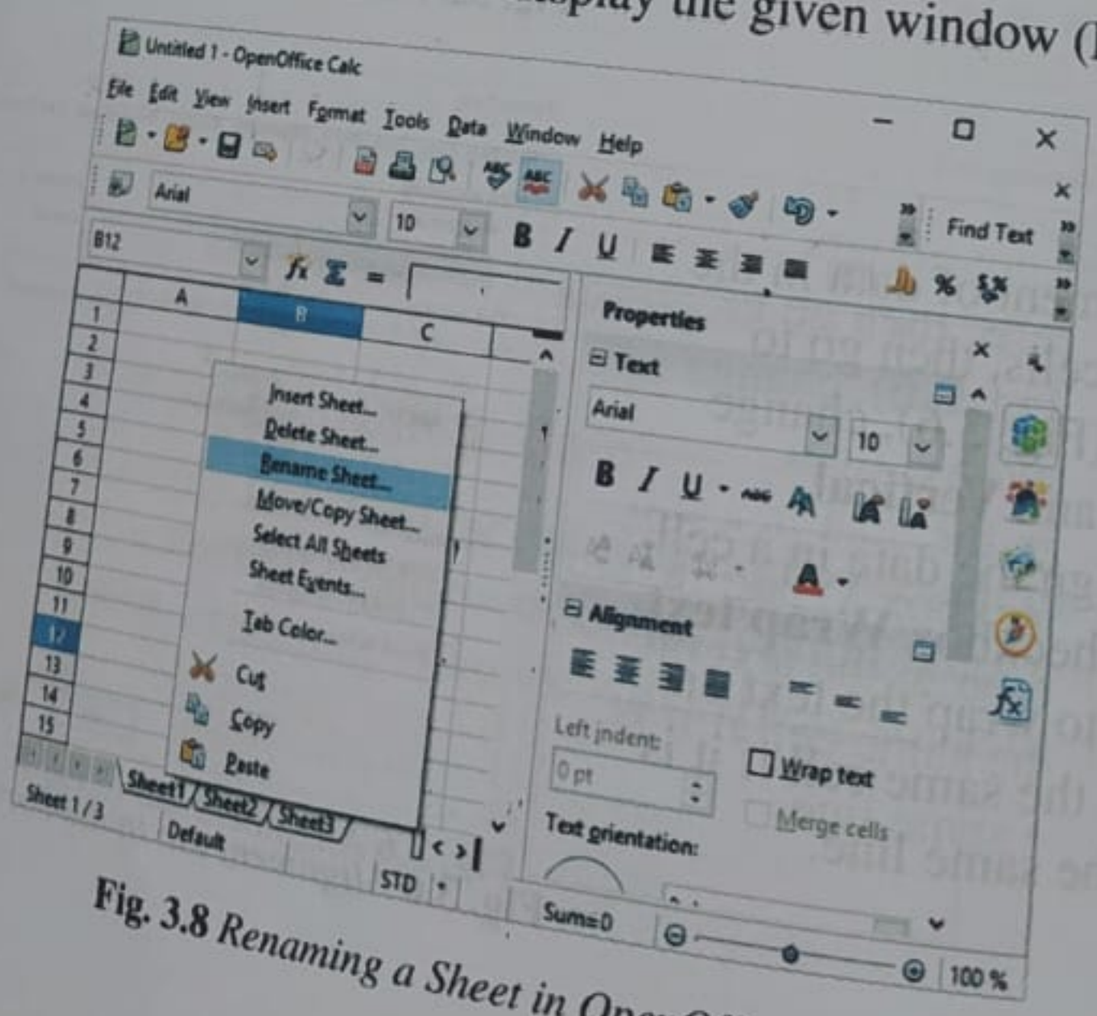


Fig. 3.8 *Renaming a Sheet in OpenOffice Calc*

3 Introduction to OpenOffice



SNAP RECAP

1. What is open source software?
2. Differentiate between open and closed source software.

LEARNING OBJECTIVES

You will learn about:

- entering a data in a cell in OpenOffice Calc
- formatting the cells in OpenOffice Calc
- renaming a worksheet in OpenOffice Calc
- writing functions in OpenOffice Calc
- some functions in OpenOffice Calc
- charts in OpenOffice Calc
- need for creating a database
- OpenOffice Base as relational database management system
- creating a new database
- database objects and elements of table

OpenOffice Calc

OpenOffice Calc is part of OpenOffice suite also used as a spreadsheet tool. It is a common name given to a computer application that stores and displays information in a tabular form. We can also use some built in features of the software to do the calculation and analysis of the data.

The layout of the Calc window (Fig. 3.1) is the same as that of other applications in the OpenOffice suite.

A file in OpenOffice Calc is referred to as a **workbook** with the default name as Untitled 1. It is saved with an extension of .ods. A workbook



Fig. 3.1 *OpenOffice Calc window*

can have multiple sheets
divided into a grid of rov
A **row** is labelled as num
A **column** is labelled as
A **cell** is formed by an in
G32, etc.
A collection of the cells

QUICK KEY

To go to first cell of t
To reach the last cell

Entering Data in a

You can fill a workbook
of text, numbers or com
numbers, it is right.

In order to enter data in

1. Select the cell where
2. Type the value that y
3. Press the Enter key



Fig. 3

1. To change the number formatting by putting currency sign in front of amount, select the cells you want to format, then go to the **Numbers** tab in the Cells… option and select **Currency** in **Category**, change **Format code** to Indian Rupees and **Decimal Places** as 2 (Fig. 3.4).

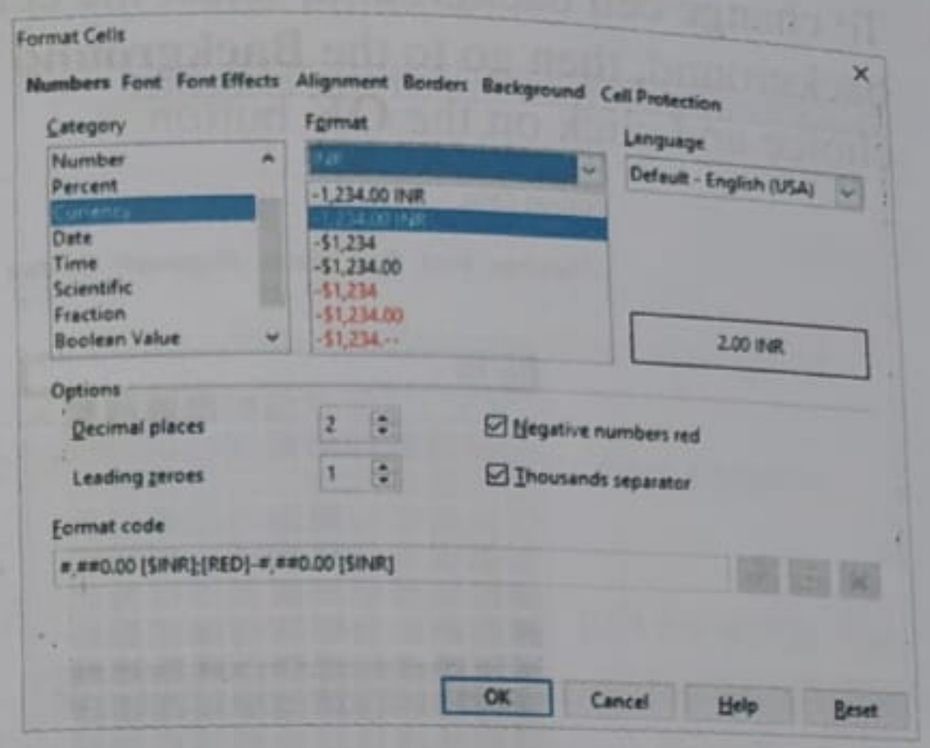


Fig. 3.4 *Numbers tab in the Format Cells dialog box*

2. To change the font name and size of the text, select the cells you want to format, then go to the **Font** tab in the Cells… option; select **Font** as Arial, **Typeface** as Regular and **Size** as 14 (Fig. 3.5).

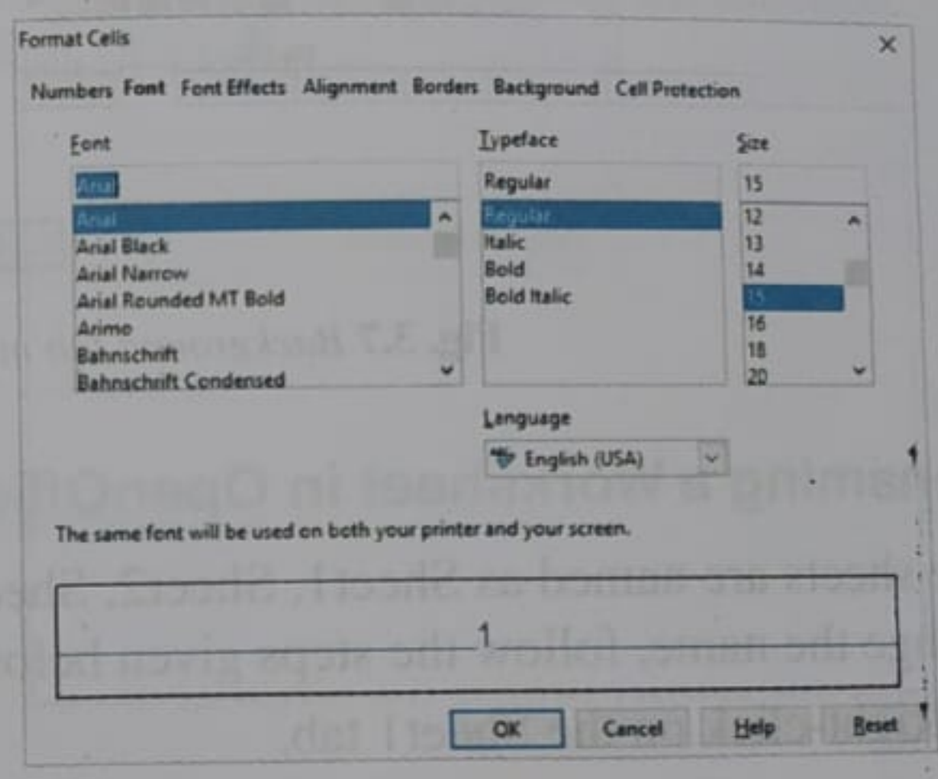


Fig. 3.5 *Font tab in the Format Cells dialog box*

3. To change alignment of data in the cells, select the cells, then go to **Alignment** tab (Fig. 3.6), change the **Horizontal** and **Vertical** alignment to align the data in a cell and select the checkbox **Wrap text automatically** to wrap the text on the next line in the same cell if it is not fitting on the same line.

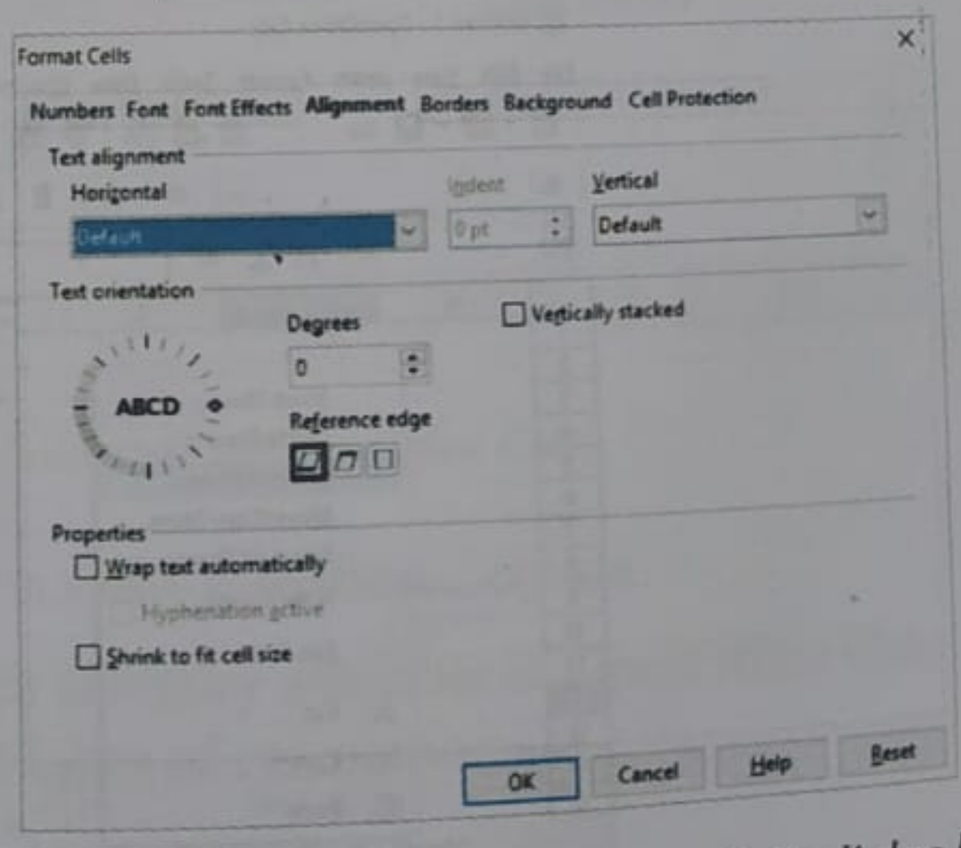


Fig. 3.6 *Alignment tab in the Format Cells dialog box*

A formula or a function can contain:

- Numbers like = 45 + 12
- Cell range = Sum (D2:D5)
- Cell address = A1 + B4
- Both = 34 + E6 + F6 + 25

When we type in a formula and press Enter, the formula will not be visible in a cell anymore. Only the result of that formula will be visible in a cell. The formula can be seen in Input Line of the Formula bar. We can edit the formula in the Input Line of the Formula toolbar when the cell is selected or we can double-click the cell. The value changes automatically with the change in the formula.

How to Use Functions in a Workbook

The step by step procedure to create a sample worksheet for result analysis of a stude is given below:

1. Create a worksheet with S.No, Marks 1, Marks 2, Marks 3 for result analysis (Fig. 3.11).

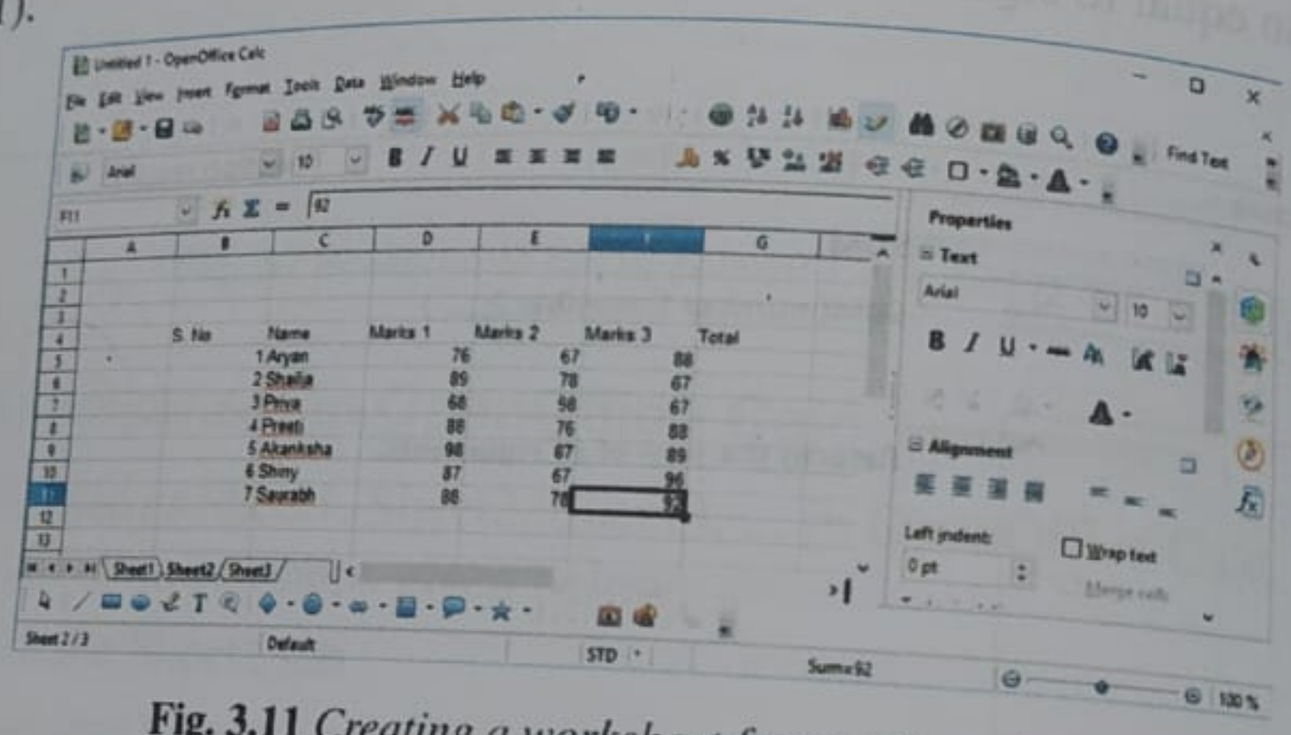


Fig. 3.11 *Creating a worksheet for result analysis*

2. Enter the formula =D5+E5+F5 or =SUM(D5:F5) in cell G5. As soon as you press the Enter key, the formula disappears to the formula bar and the result appears in the cell (Fig. 3.12).

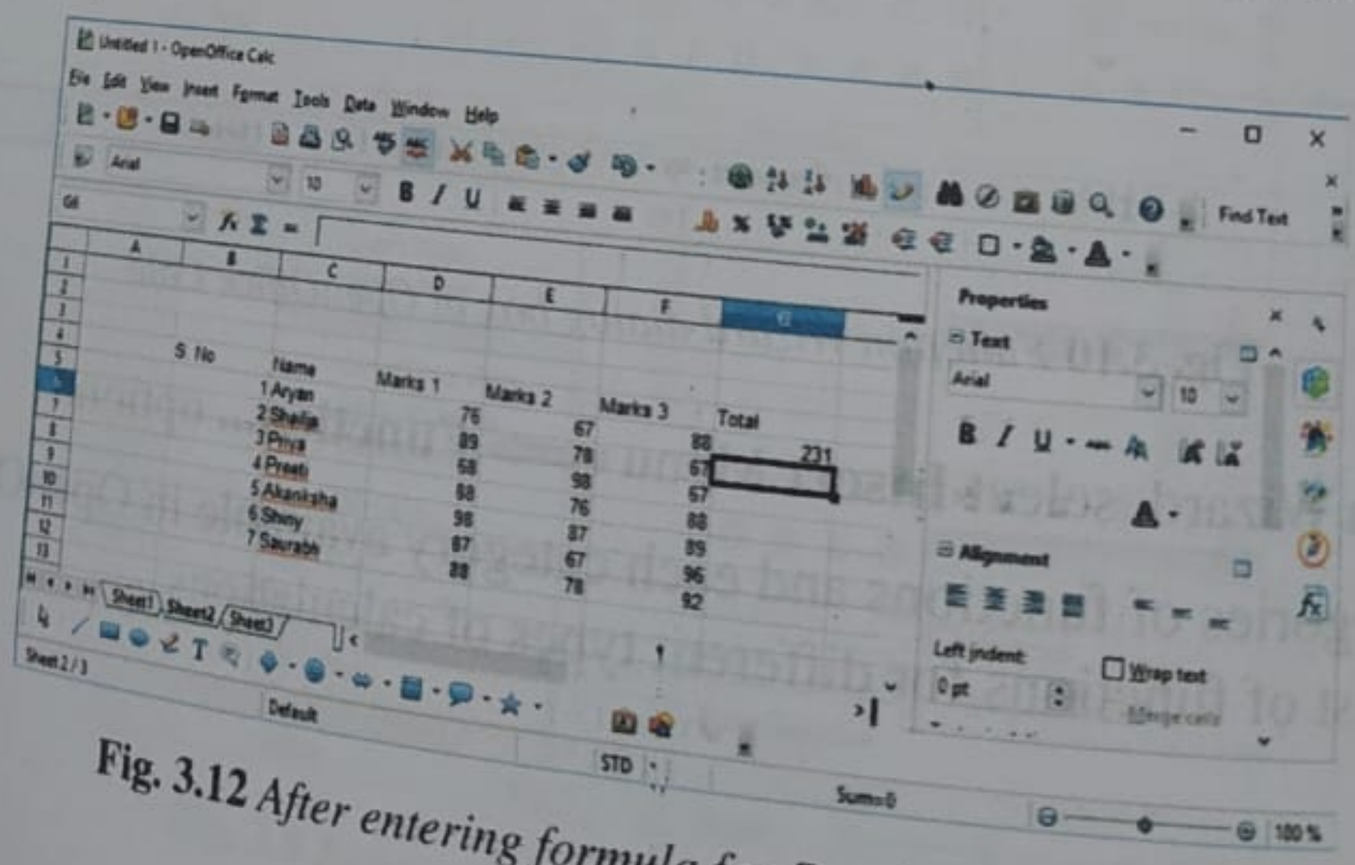


Fig. 3.12 *After entering formula for Sum in a worksheet*

3. The **Rename Sheet** dialog box will appear as shown in Fig. 3.9. Change the **Name** say to **MyFriends** and click on *OK*.

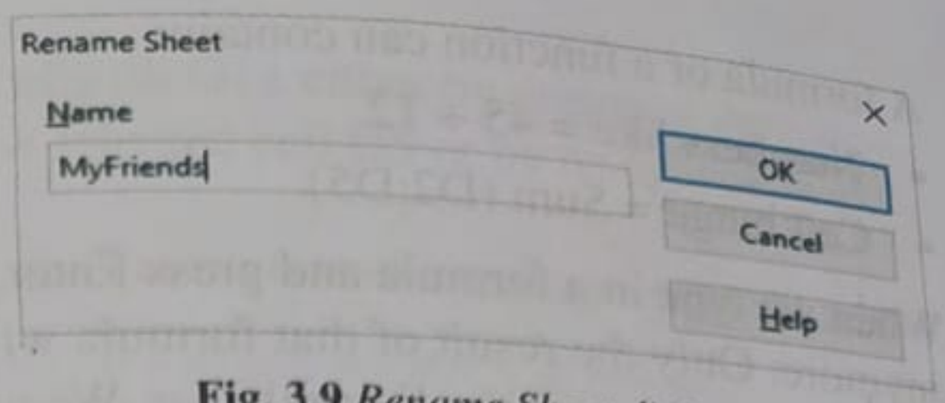


Fig. 3.9 *Rename Sheet dialog box*

Writing Functions in OpenOffice Calc

A formula is an expression that begins with an equal to (=) sign. You can enter a formula in a cell to do calculations on data.

Sometimes you may not remember a formula or it may be a tedious job to write the addition formula for 100 rows; in that case, you may use built-in formulas available in OpenOffice Calc to perform calculations. These built in formulas are called **Functions**.

A function can be entered either through Function Wizard or by typing the function beginning with an equal to sign with the values in a cell.

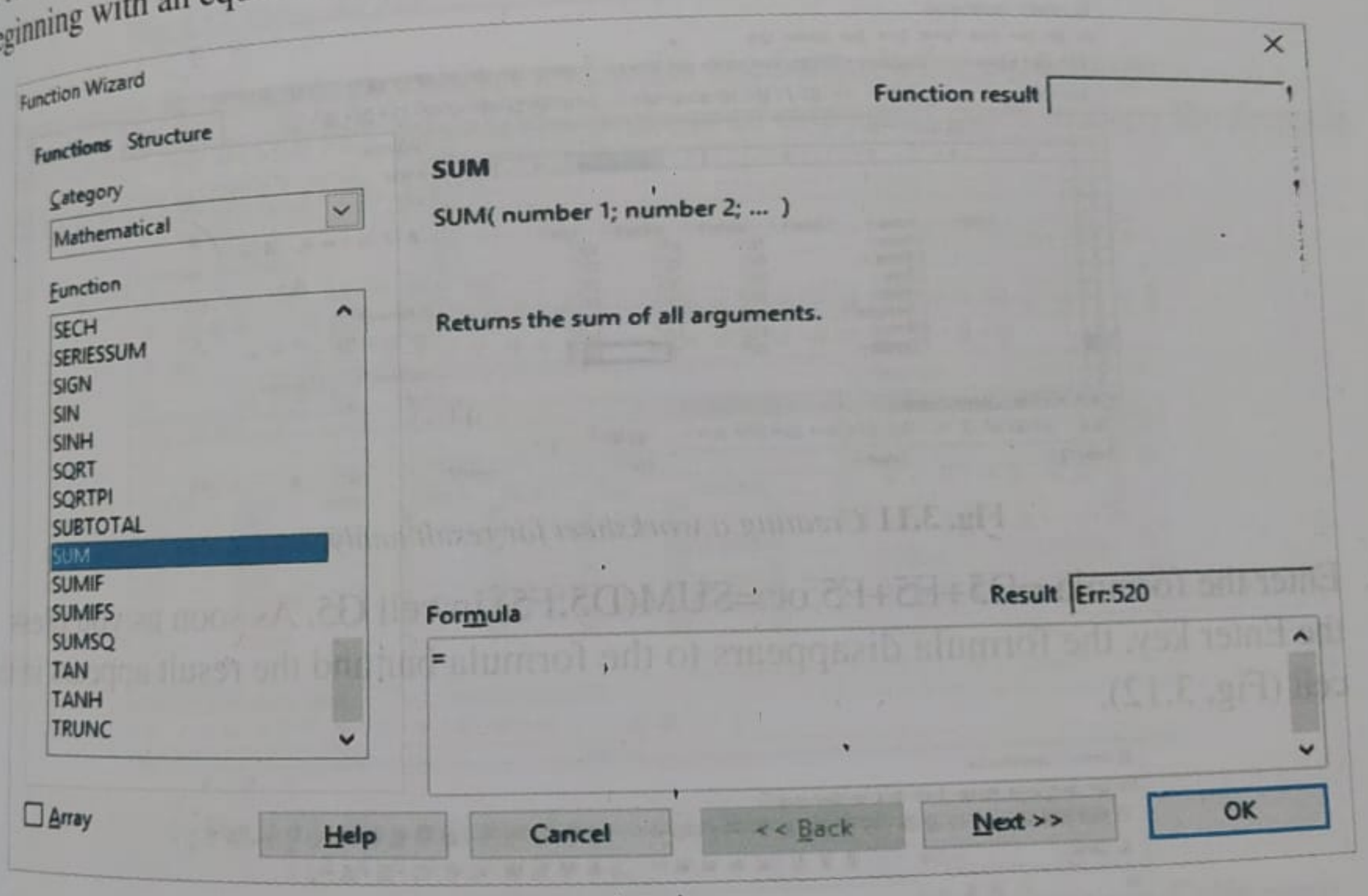


Fig. 3.10 *Function Wizard dialog box in OpenOffice Calc*

To open Function Wizard, select **Insert** Menu ⟹ **Function...** option.

There are 10 categories of functions and each category available in OpenOffice Calc includes a long list of functions for different types of calculations.

7. After all the calculations are done, the spreadsheet will appear as the one shown in Fig. 3.15.

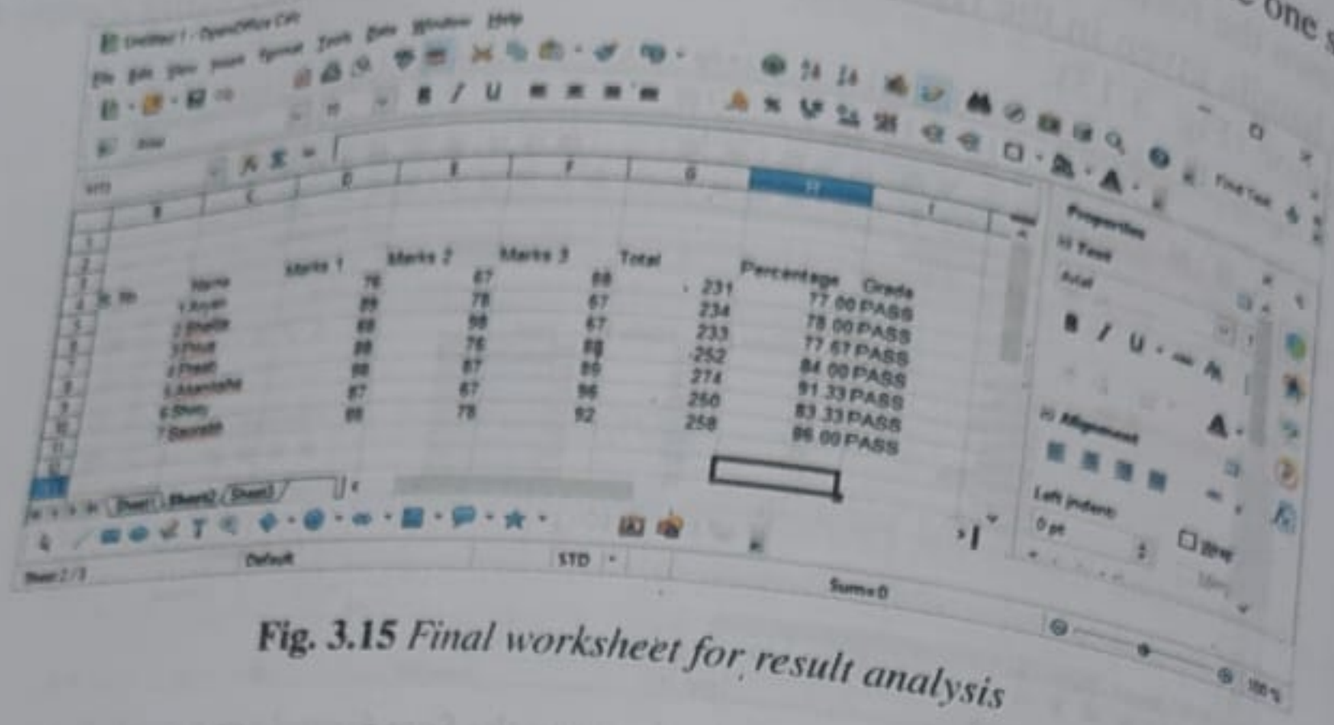


Fig. 3.15 *Final worksheet for result analysis*

Some Functions in OpenOffice Calc

Some of the commonly used functions with examples are given below.

- **SUM()**: It adds up the values given as parameters in the function. The parameters are separated by a semicolon (;) in between. These values can be the cell address, cell range or number.

  For example,

  =SUM(D5;E5;F5) will give the answer 231

  OR

  =SUM(D5:F5;D12)

- **MAX()**: It returns the highest value from the list of the values given as parameters in the function.

  For example, to find out the highest marks in Marks1 in the list created above, we write:

  =MAX (D5:D11) will give the answer 98

  OR

  =MAX (D5;D6;D7)

- **MIN()**: It returns the lowest value from the list of the values given as parameters in the function.

  For example, to find out the lowest marks in Marks3 in the list created above. We

=MIN(F5:F11) will g

=MIN(F5;F6;F7)

- **AVERAGE()**: It calc function.

  For example, to find =AVERAGE(D5:D11

- **IF()**: It evaluates the then the second param evaluated.

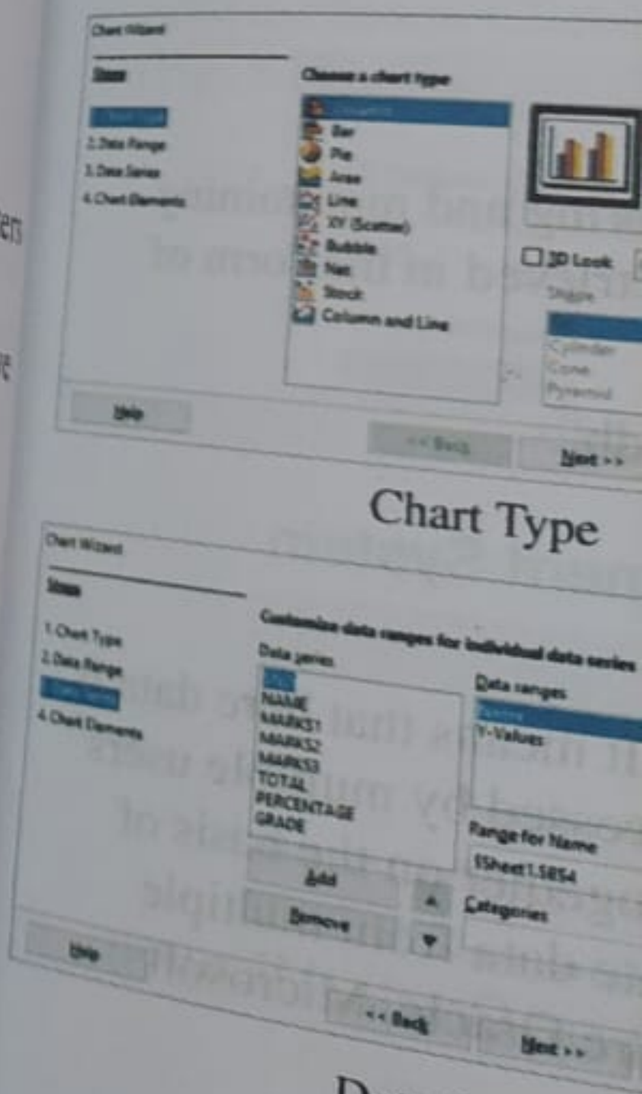
  For example, to displa worksheet will be:

  =IF(H5>40, "PASS","

Charts in OpenOffic

Charts are graphical repre the existing data by using **Insert** menu ⟹ **Chart.**

It will display the Chart w four steps which are about



Fi

=MIN(F5:F11) will give the answer 67

OR

=MIN(F5;F6;F7)

- **AVERAGE()**: It calculates the average of the values given as parameters in the function.

For example, to find out the average of Marks2, use the formula:

=AVERAGE(D5:D11)

- **IF()**: It evaluates the condition given as the first parameter. If the condition is true, then the second parameter will be evaluated, otherwise the third parameter will be evaluated.

For example, to display PASS or FAIL based on the condition given for the above worksheet will be:

=IF(H5>40, "PASS","FAIL")

Charts in OpenOffice Calc

Charts are graphical representation of data in a spreadsheet. You can create a chart of the existing data by using the Chart tool present on the toolbar or by selecting **Insert** menu ⟹ **Chart...** option.

It will display the Chart wizard. In order to create a complete chart, you need to follow four steps which are about choosing the following things as shown in Fig. 3.16.

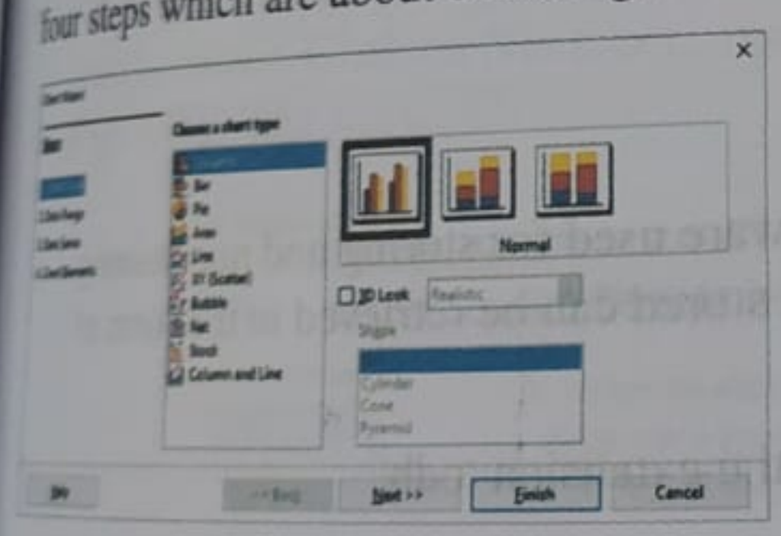
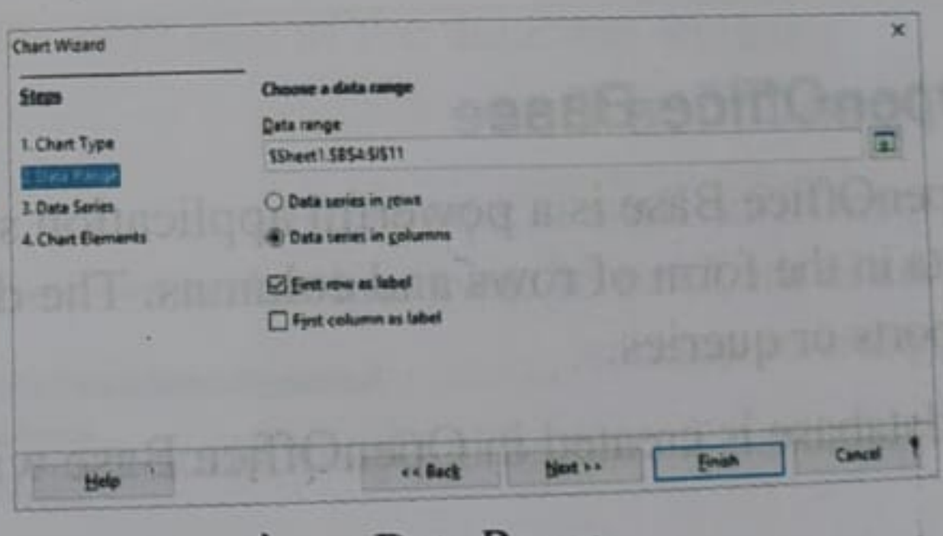
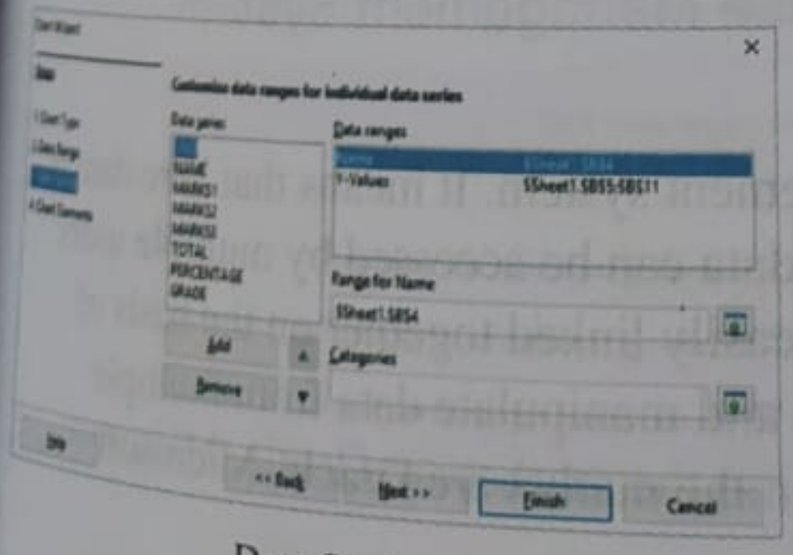


Chart Type



Data Range



Data Series

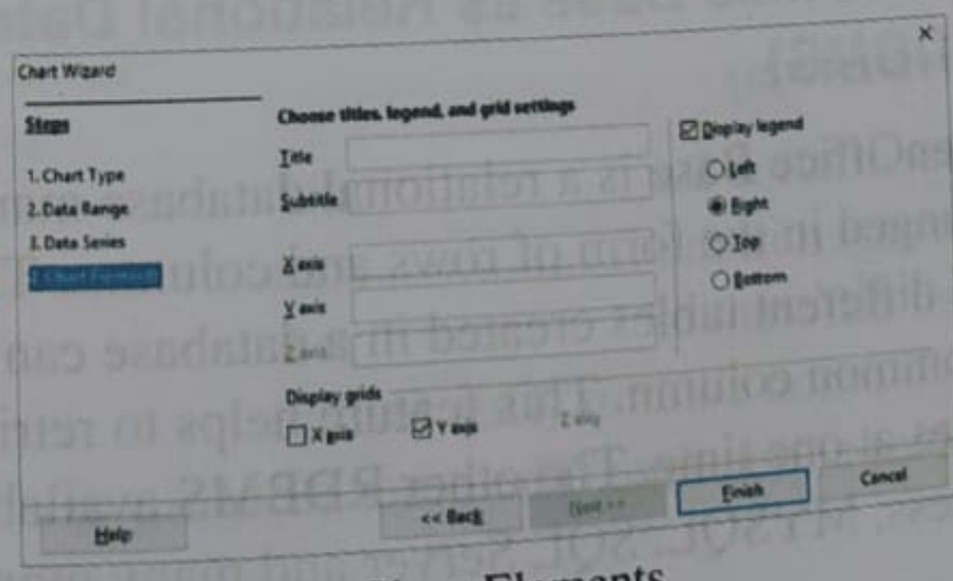


Chart Elements

Fig. 3.16 *Chart Wizard in OpenOffice Calc*



ACTIVITY

Complete the following activity.

1. *Create a list of shopping items purchased from a shopping mall with the following columns – Item code, Name, Quantity purchased and Rate of each item.*
2. *Make a new column* **Amount per item** *to calculate the amount of each item; use the formula =Rate*Quantity.*
3. *Make a list of 10 items purchased and calculate the Total Billing Amount. If the Billing Amount is more than ₹ 10,000/- then give a discount of 15% or 5%.*
4. *Calculate the actual bill after deducting the discount.*

TRY THIS



Save the above created items file by selecting **File ⟹ Save As... ⟹** tick the checkbox **Save with password**. **Set password** to open file ⟹ OK.

Now close the file and try to open it. What will happen?

Find out if you can save a file with permission to open and view but cannot modify the content.

OpenOffice Base

OpenOffice Base is a powerful application software used for storing and maintaining data in the form of rows and columns. The data stored can be retrieved in the form of reports or queries.

A database is created in OpenOffice Base with the extension .odb.

OpenOffice Base as Relational Database Management System (RDBMS)

OpenOffice Base is a relational database management system. It means that here data is arranged in the form of rows and columns. This data can be accessed by multiple users and different tables created in a database can be easily linked together on the basis of a common column. This feature helps to retrieve and manipulate data from multiple tables at one time. The other RDBMS available in the market are Oracle, Microsoft Access, MYSQL, SQL Server and many more.

3. *Copy down the formula in the rest of the cells till G11 either by dragging the autofill handle given in the right side of the selected cell G5 or by using a copy and paste option (Fig. 3.13).*

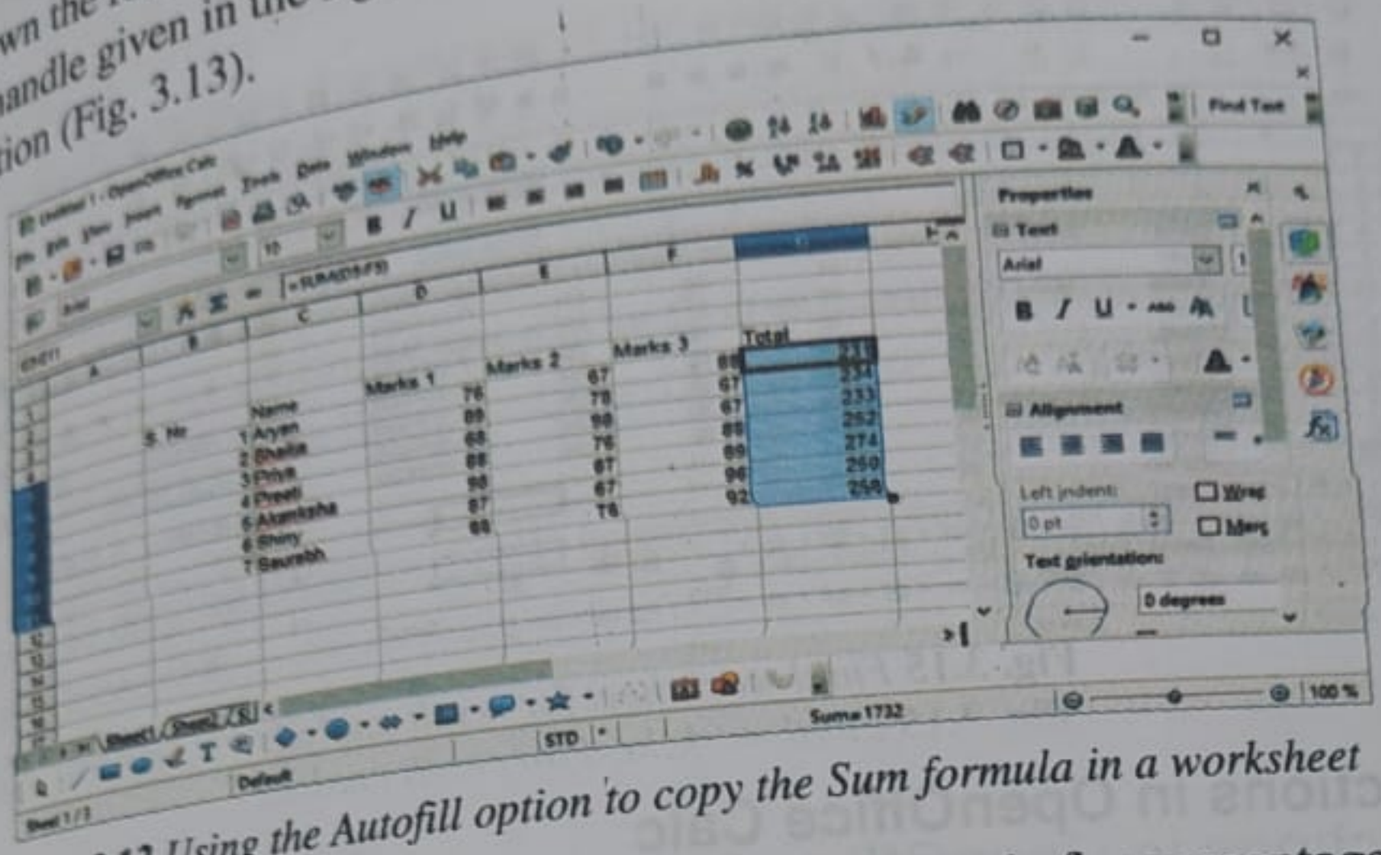


Fig. 3.13 *Using the Autofill option to copy the Sum formula in a worksheet*

4. Next, insert the percentage column and write the formula =(G5/300)*100 in cell H5. Repeat the process of copy and paste to copy the formula from cell H5 till H11 (Fig. 3.14).

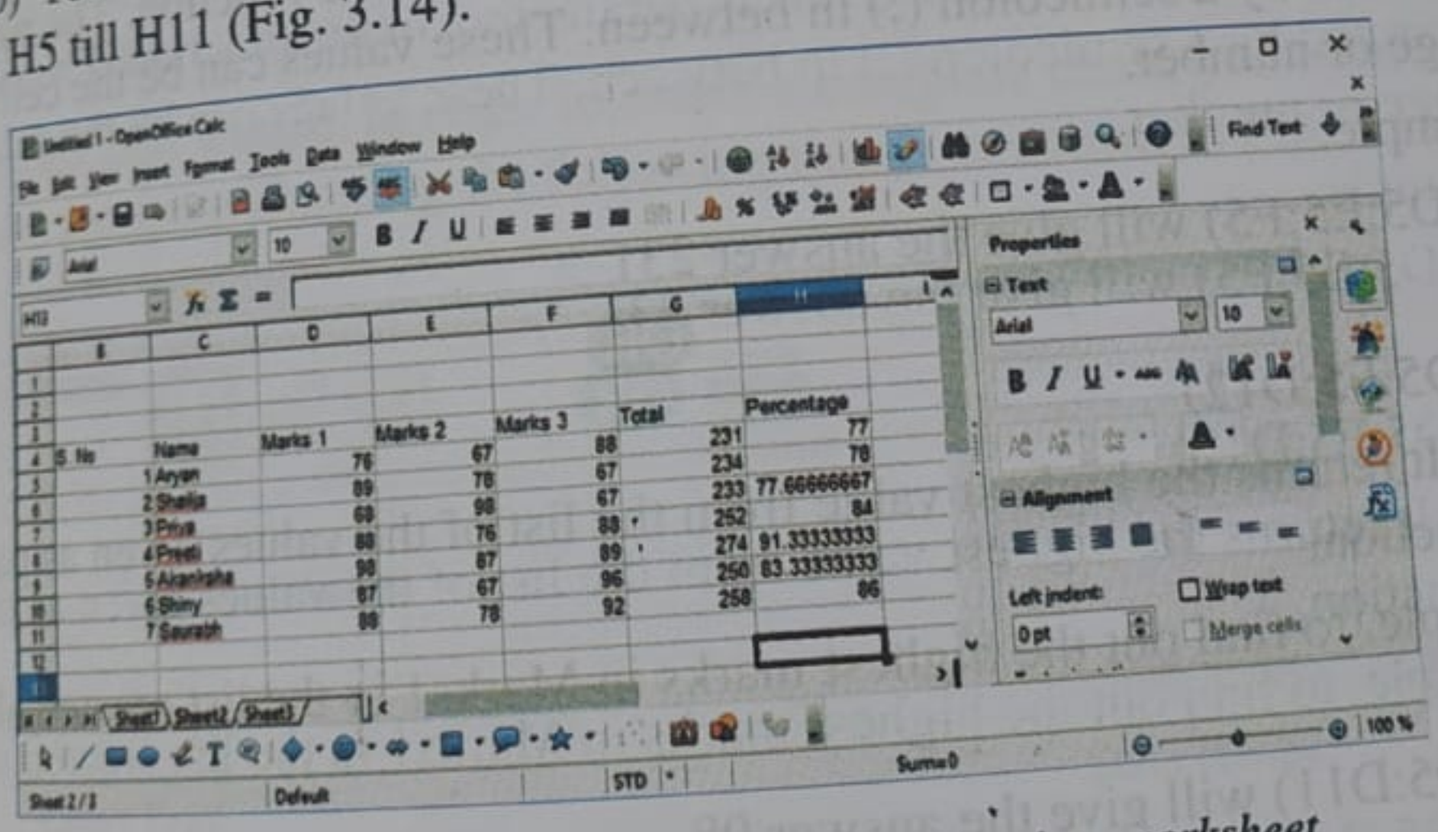


Fig. 3.14 *After entering formula for Percentage in a worksheet*

5. Format the number up to 2 decimal places by selecting **Format** ⟹ **Cells** ⟹ **Numbers** ⟹ **Category: Number.**

OR

By selecting the **Number Format: Delete Decimal Place** tool present on the **Toolbar.**

6. Now insert the Grade column and write the formula =IF(H5 > 40;"PASS";"FAIL") in cell I5 to display PASS or FAIL on the basis of the percentage calculated.

5. Now click on **Tables** object. The **Tasks** will show three different ways of creating tables in OpenOffice base. Select **Create table in Design View...** It will open **Table Design** window (Fig. 3.21) where we design the structure of a table.

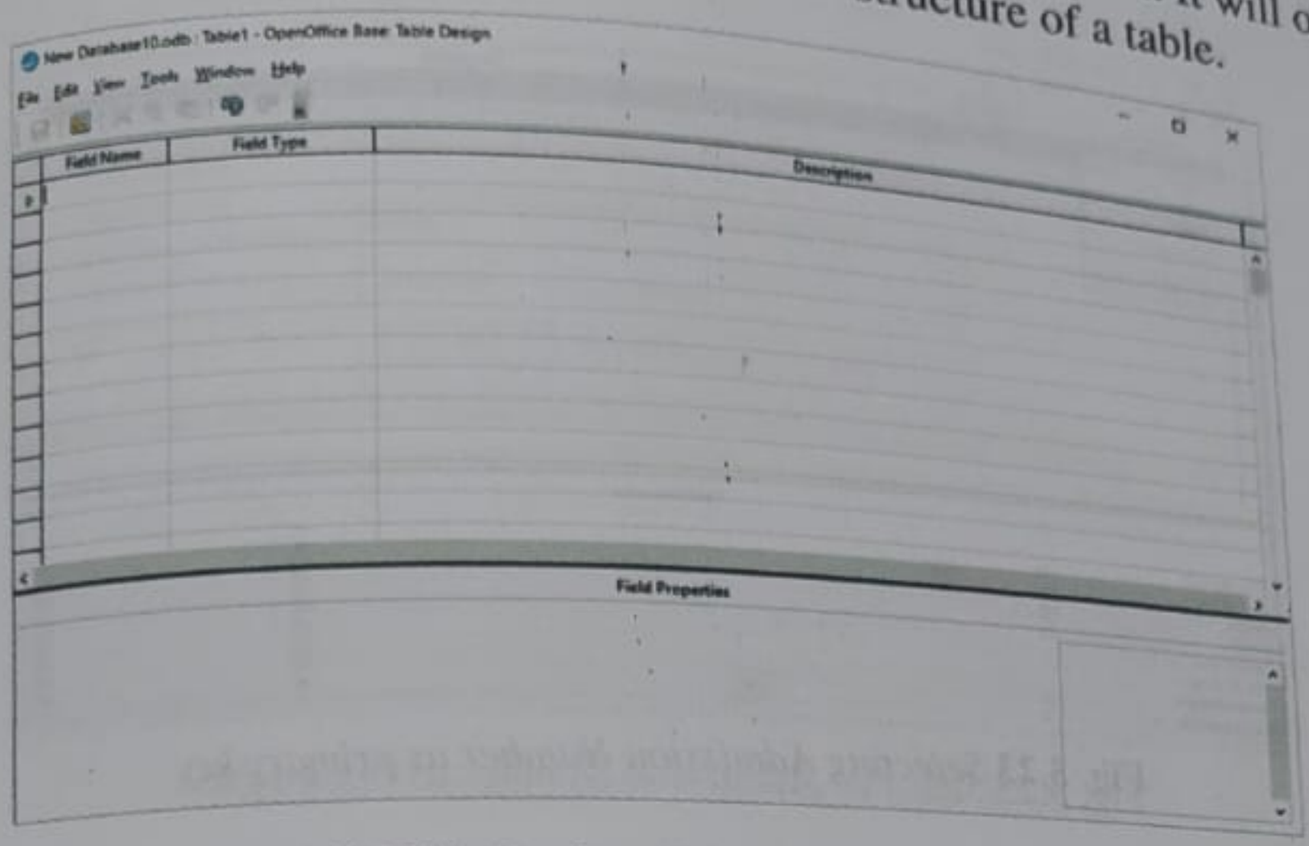


Fig. 3.21 *Table Design window of OpenOffice Base*

6. Let us now design a **Student table** (Fig. 3.22). In the **Field Name** type **Admission Number** and **Field Type** will display the list of different data types with Text [VARCHAR] as default in the drop-down list. Select Text [VARCHAR]. Select **Description** to write a few lines about the description of a column created. Giving description of a column is optional.

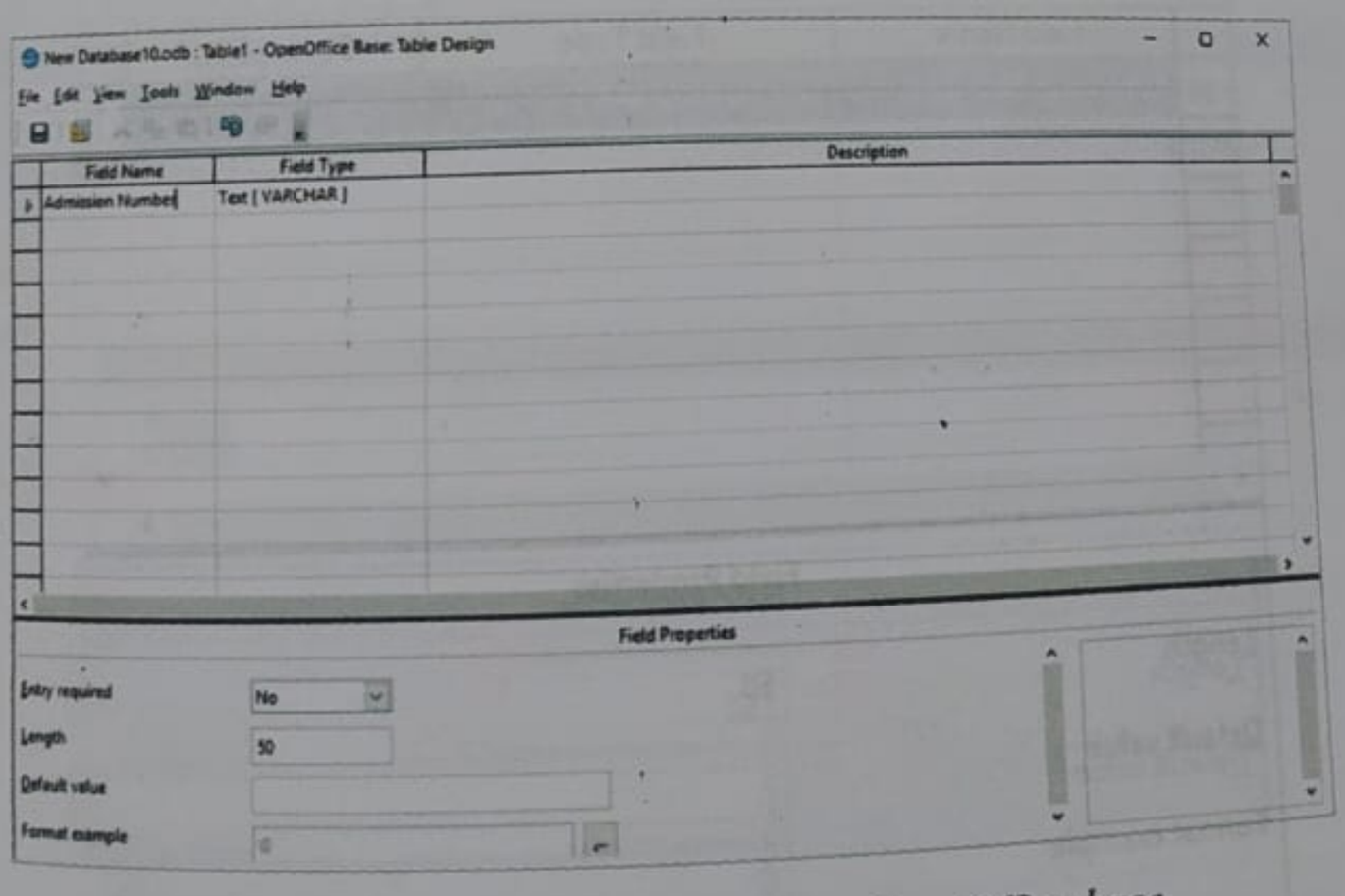


Fig. 3.22 *Creating Student table in OpenOffice base*

Creating a New Database

1. Click on **File** ⟹ **New** ⟹ **Database** to create a new database. A **Database Wizard** will appear as shown in Fig. 3.17.

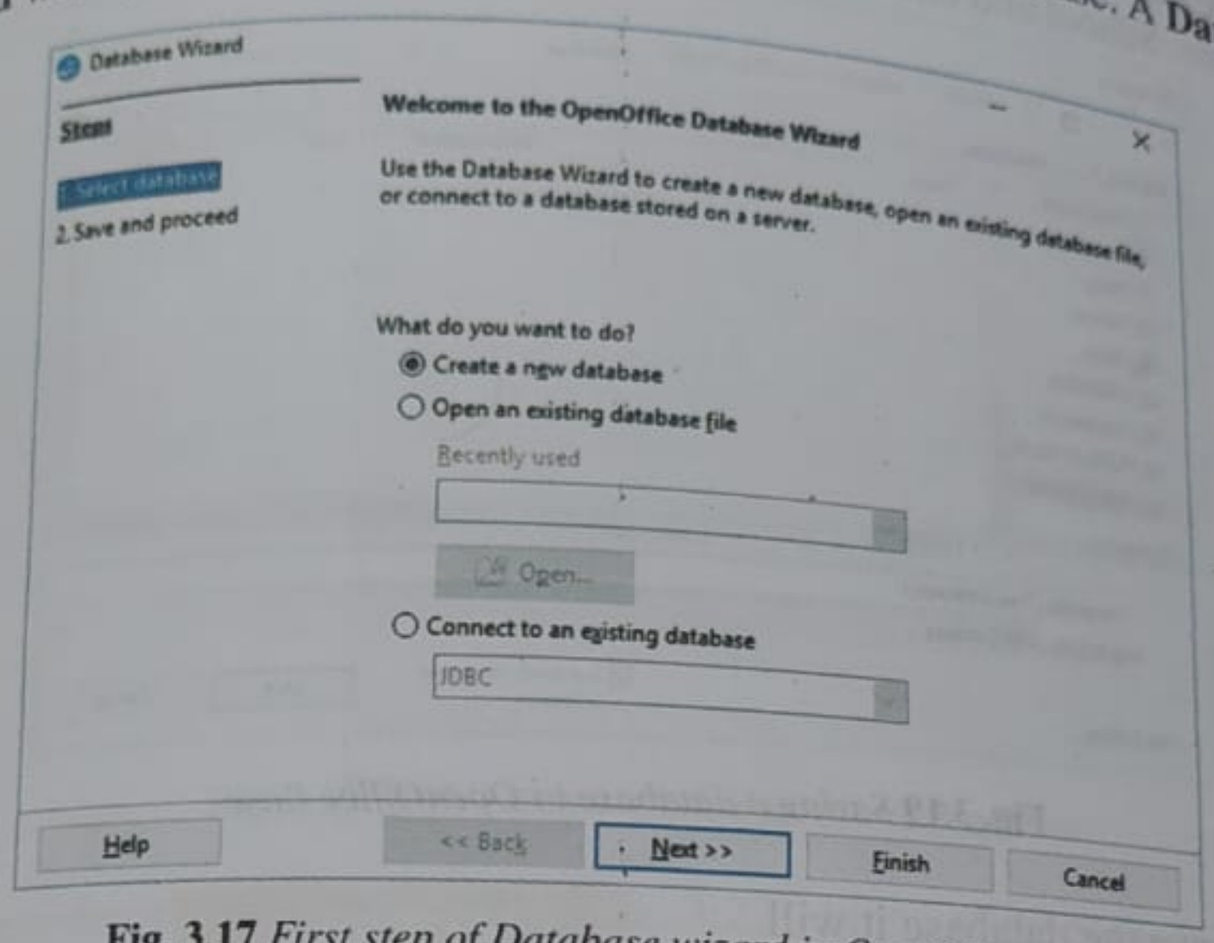


Fig. 3.17 *First step of Database wizard in OpenOffice Base*

2. Select **Create a new database** if a new database is required to be created, or select **Open an existing database file** if the database has been created earlier and you need to open it. Click on **Next>>** to open the next step of the database wizard (Fig. 3.18).

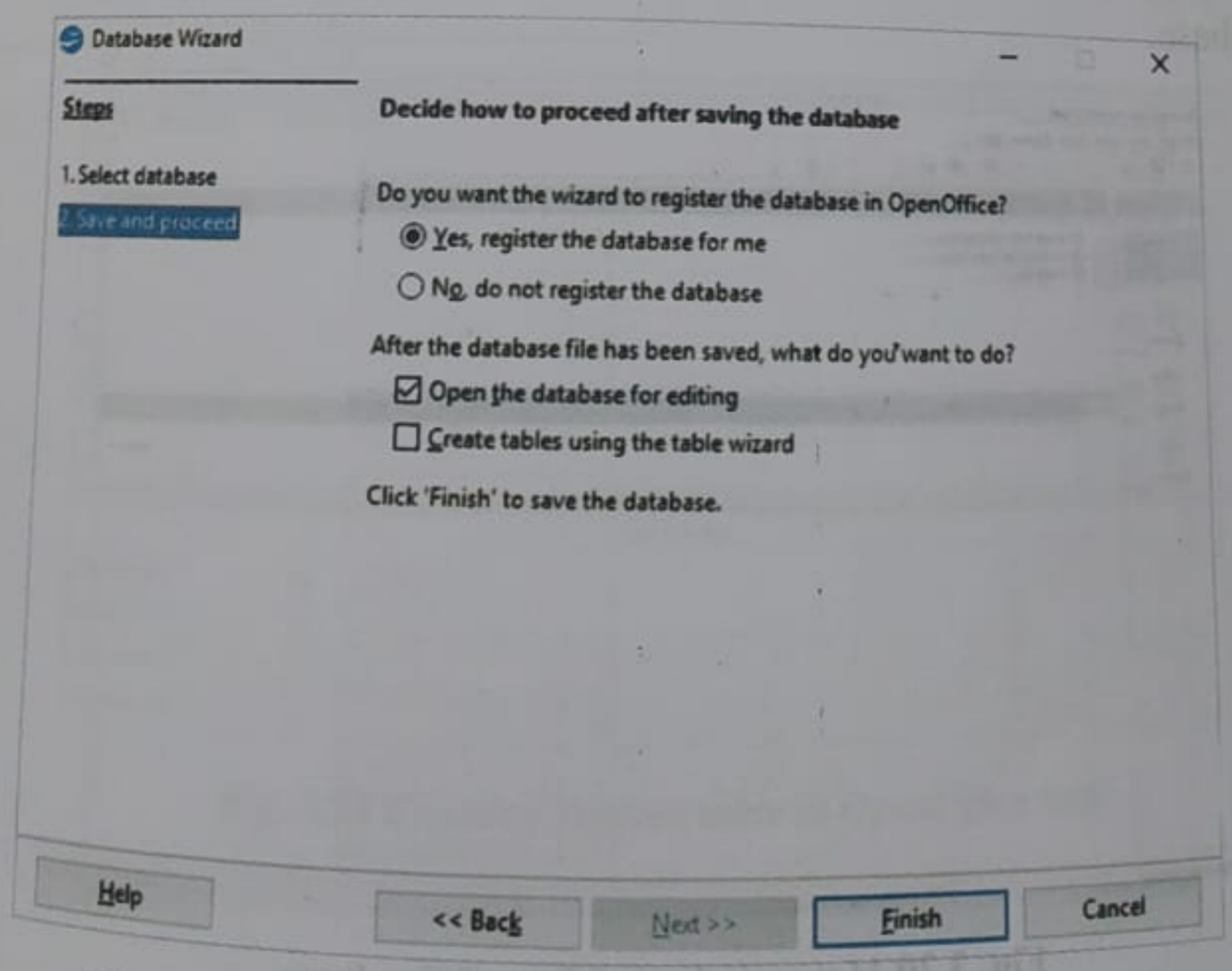


Fig. 3.18 *Second step of Database wizard in OpenOffice Base*

3. Select **Yes, register the database for me** and tick **Open the database for editing** option. Click on **Finish**. This will open the **Save As** window (Fig. 3.19). Assign a new name: School and click on **Save**.

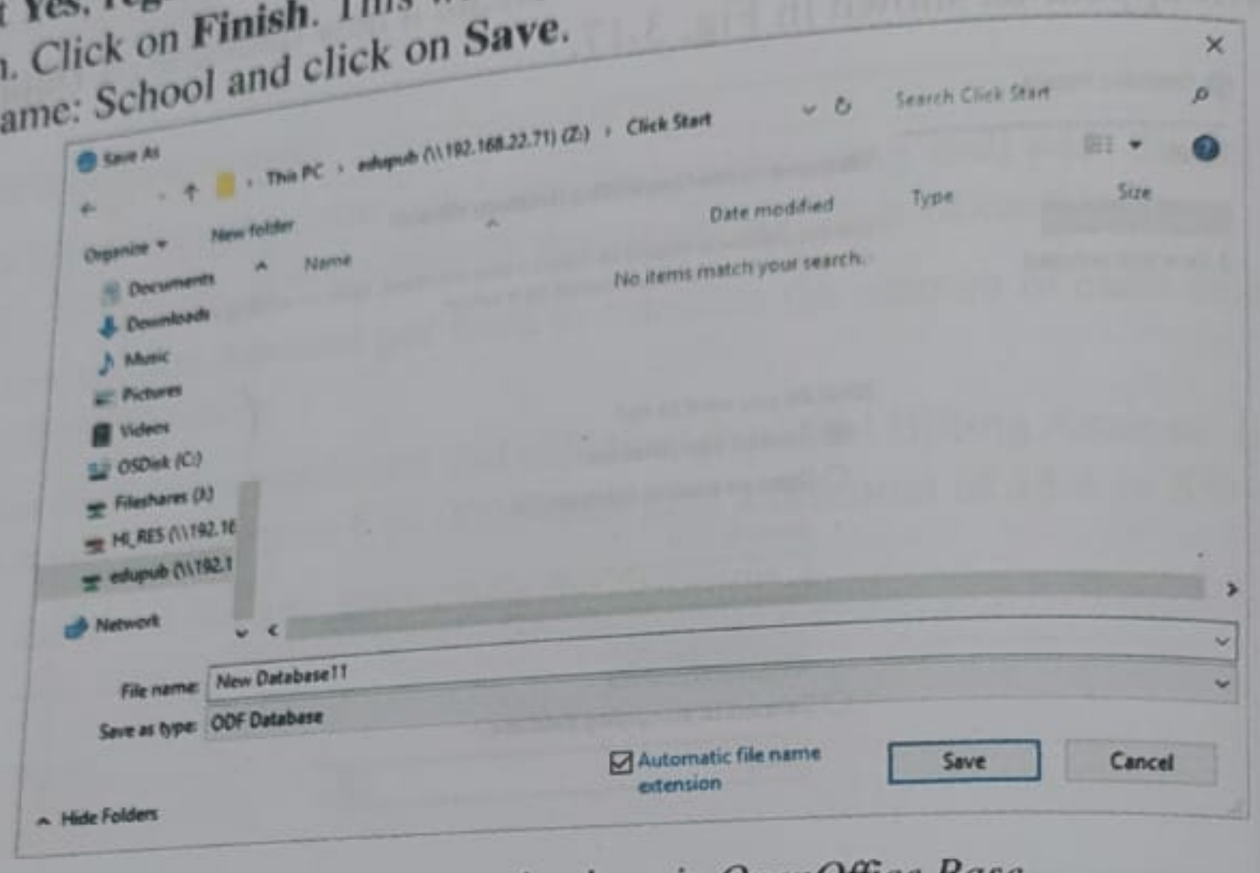


Fig. 3.19 *Saving a database in OpenOffice Base*

4. After you save the database it will display main window of OpenOffice Base (Fig. 3.20). This window is divided into three parts—**Database Tables, Tasks** related to each object, and **Object Instances** created in each database.

FACT FILE

All the files created in OpenOffice suite are saved as ODF (Open Document Format) document.



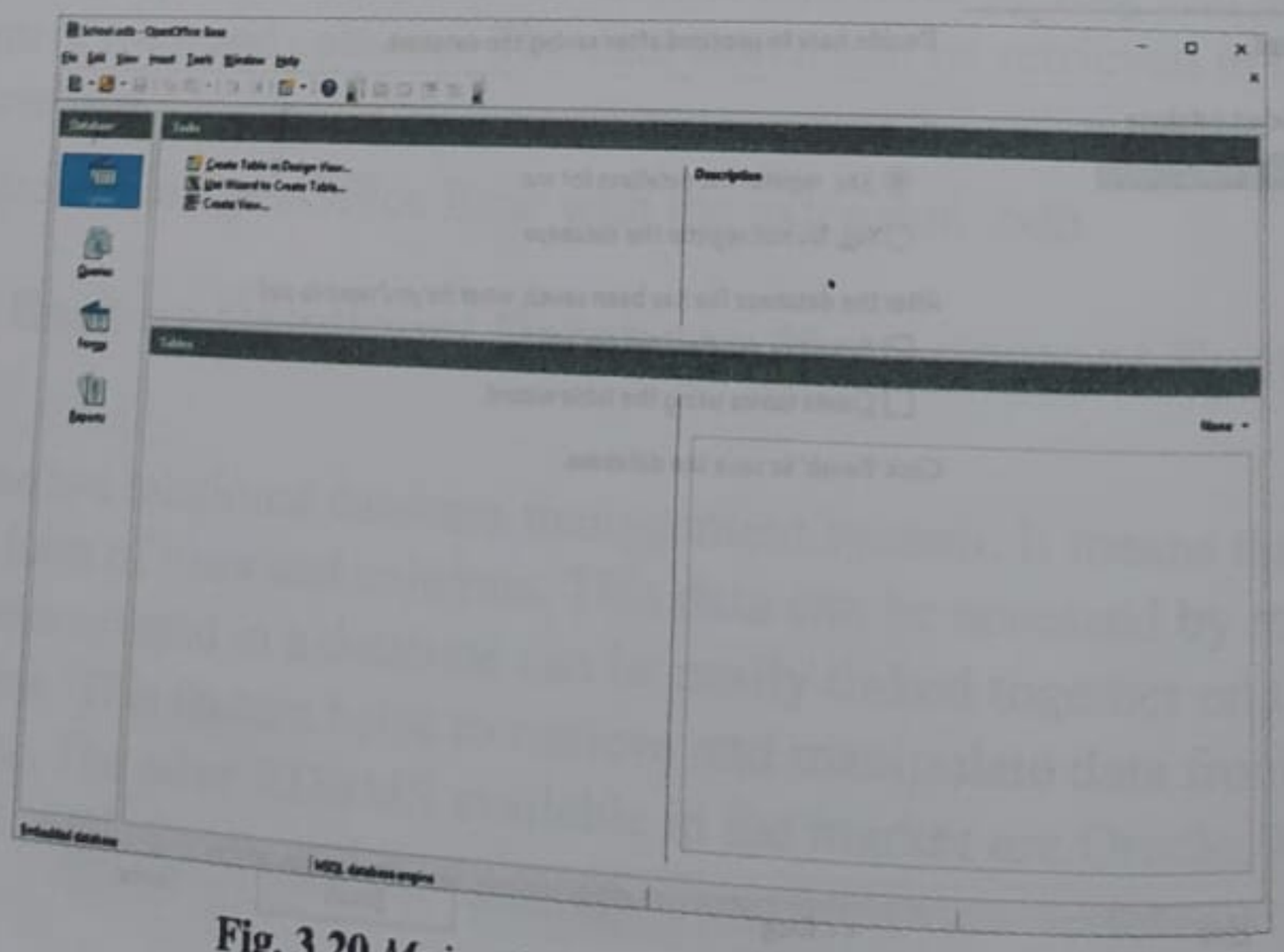
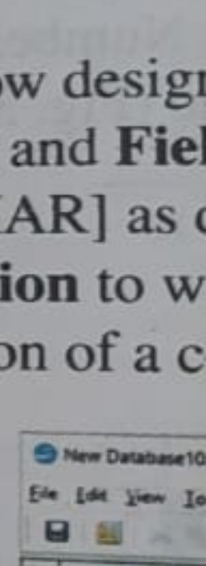
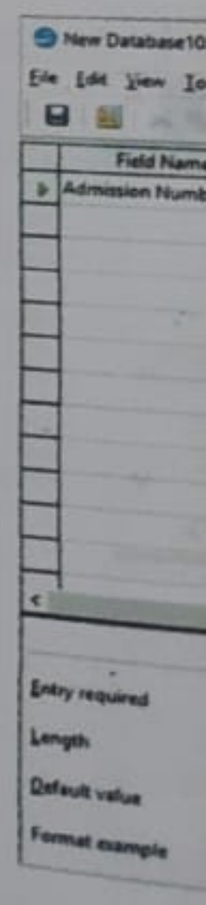


Fig. 3.20 *Main window of OpenOffice Base*

5. Now click on **Tab** tables in OpenOff **Design** window (F



6. Let us now desig **Number** and **Fiel** [VARCHAR] as **Description** to w description of a c



9. Now make few other columns with different data types in **Table Design** as shown in Fig. 3.25 and save the table with the name **Student** and click on **OK**.

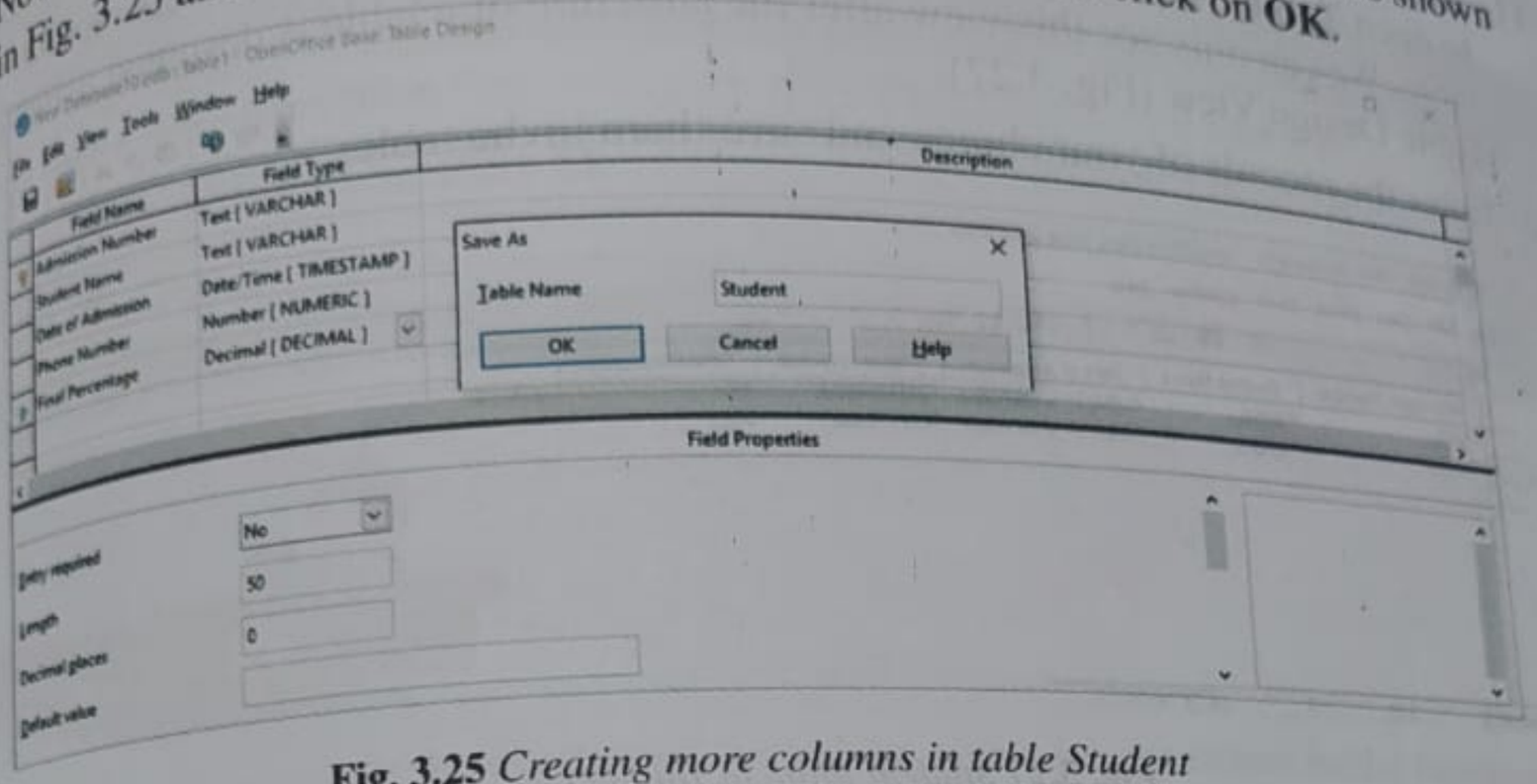


Fig. 3.25 *Creating more columns in table Student*

10. Close this file by selecting **File** ⟹ **Close** to display the window shown in Fig. 3.26. Repeat the above process to make one more table: Library. The structure of the table can be easily modified anytime by adding a new column anywhere in the structure, deleting any column and changing the existing column in the Table Design View.

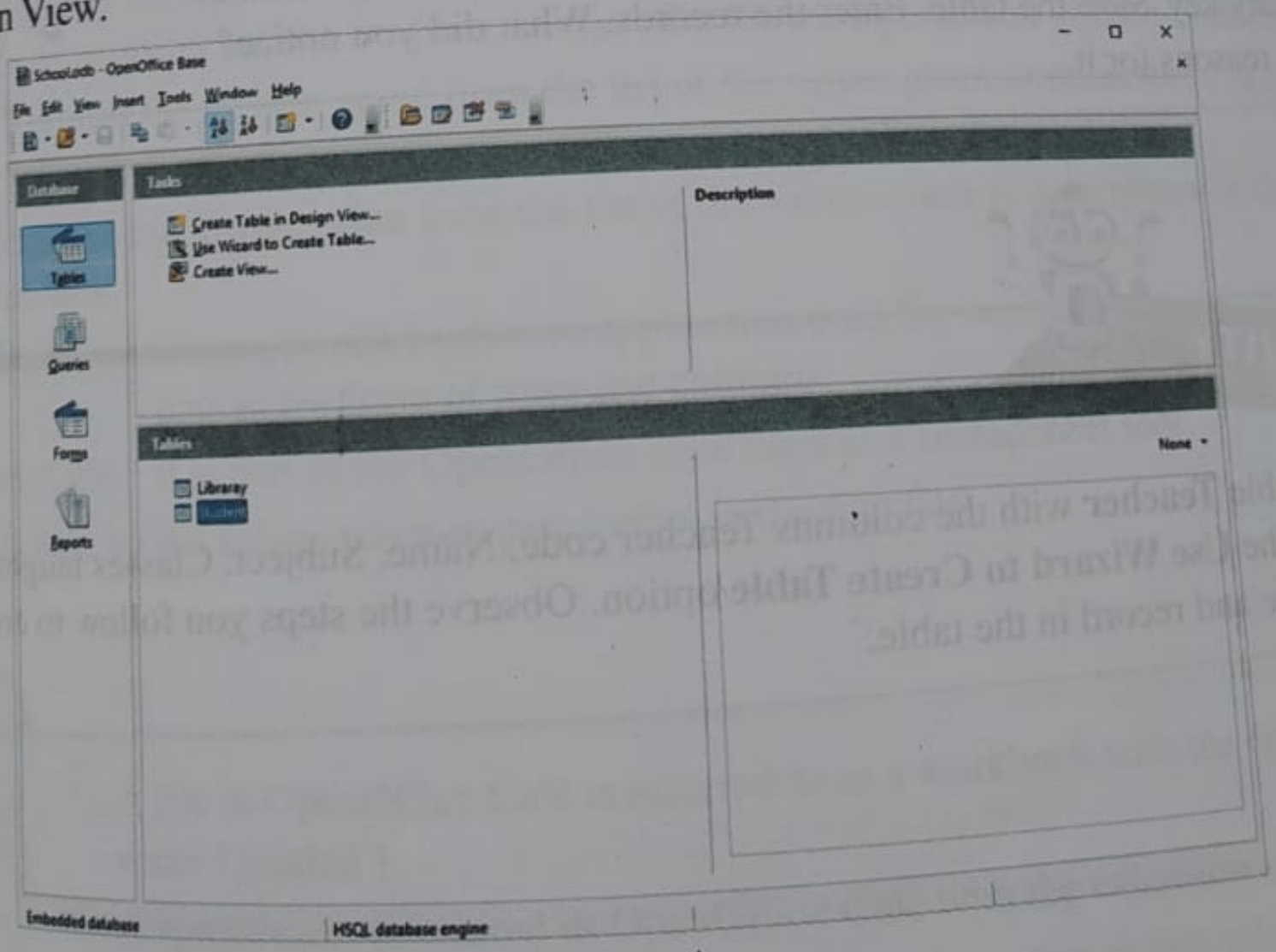


Fig. 3.26 *Creating Library table*

11. Either select **Open** from the Shortcut menu or double-click on the **Student Table** to open **Table Data View**. This view is used to add, modify and delete records in a table. We can only see this view after the structure of a table is designed using the Table Design View (Fig. 3.27).
12. Enter the records of your choice and save them in the table.

Student - New Database10 - OpenOffice Base: Table Data View

File Edit View Insert Tools Window Help

Admission Number	Student Name	Date of Admission	Phone Number	Final Percentage
V1001	Advika	01/04/19 12:00 AM	7278370000	94
V1002	Arshia	02/04/19 12:00 AM	7277370000	92

Record 2 of 2

Fig. 3.27 *Student table in Table Data view*

TRY THIS

At the time of designing the table structure, do not select a column as primary key. Save the table. Enter the records. What did you notice? Give reasons for it.





ACTIVITY

Create a table **Teacher** with the columns Teacher code, Name, Subject, Classes taught by, etc. using the **Use Wizard to Create Table** option. Observe the steps you follow to create the structure and record in the table.



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7. Right-click on the green arrow key in front of the Admission Number. This will open a shortcut menu as shown in Fig. 3.23.

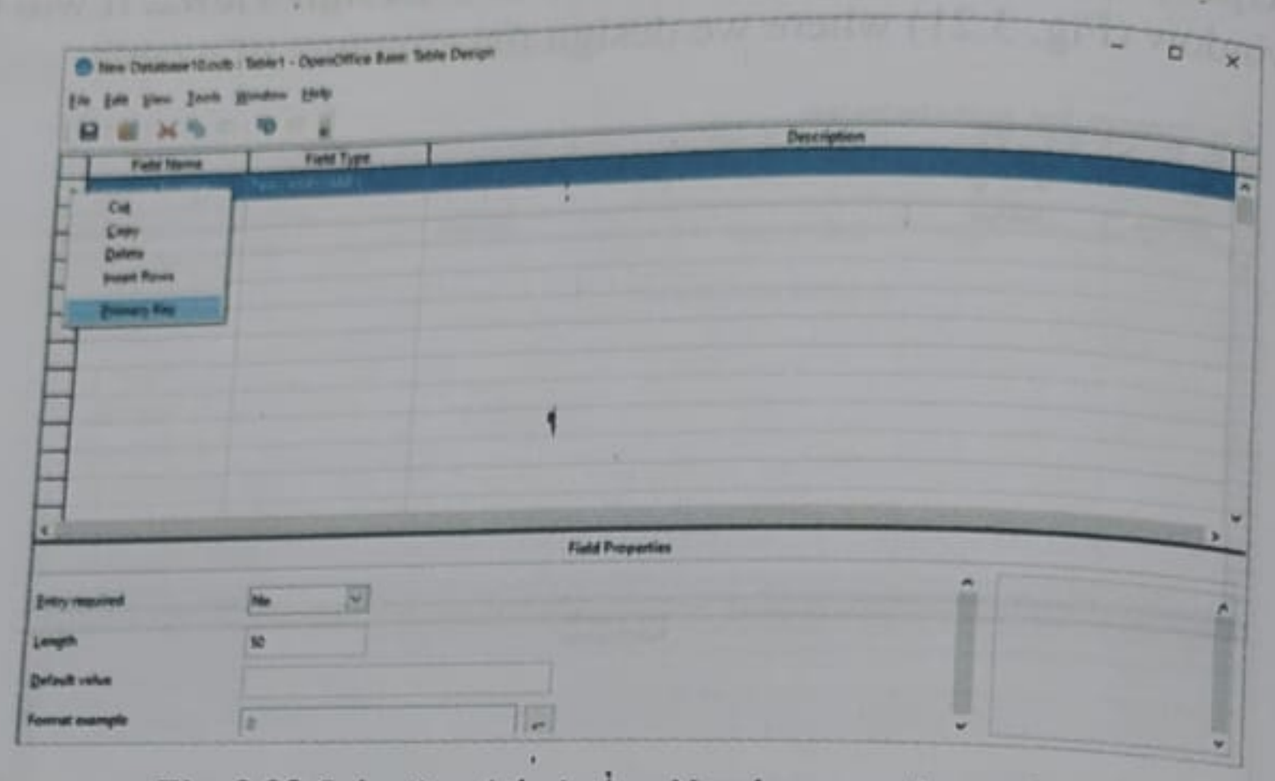


Fig. 3.23 *Selecting Admission Number as primary key*

8. Select the option **Primary Key** from the menu to make Admission Number as the primary key. A small key appears infront of the Admission Number column to show that this has been selected to save unique values in a table (Fig. 3.24).

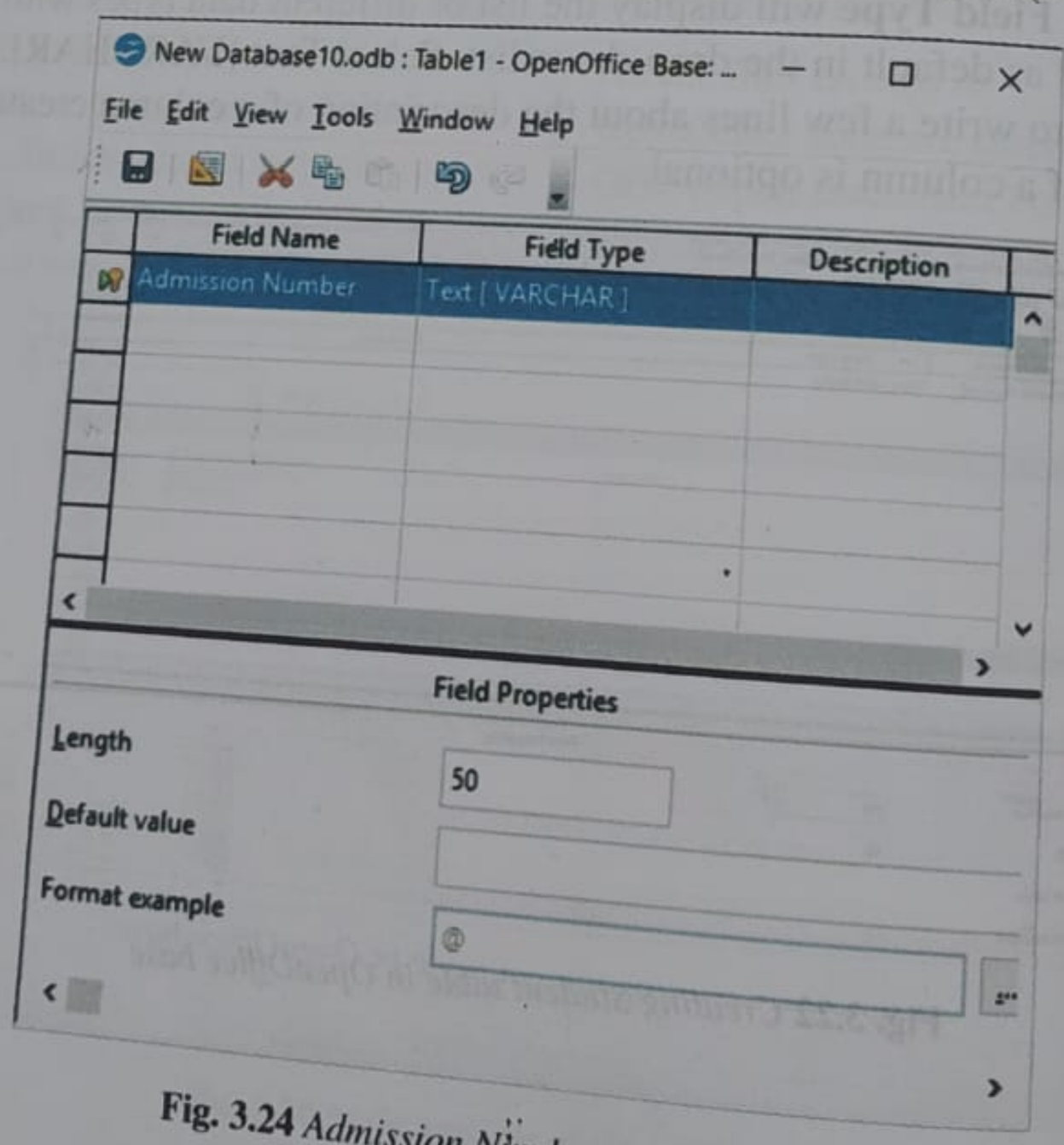
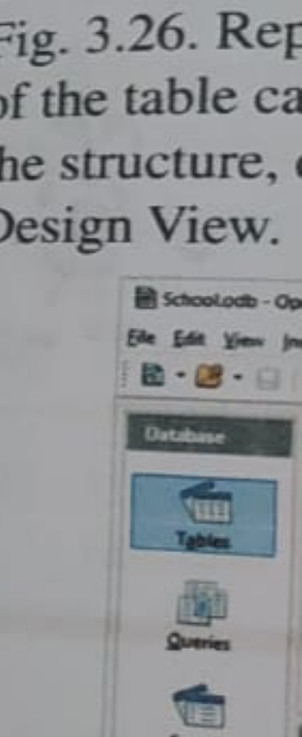


Fig. 3.24 *Admission Number as primary key*

9. Now make fe
in Fig. 3.25 a



10. Close this file
Fig. 3.26. Rep
of the table ca
the structure,
Design View.

4. A function can be entered either through Function Wizard or by typing the function beginning with an equal sign with the values in a cell.
5. Charts are graphical representation of data in a spreadsheet. You can create a chart of existing data by using the Chart tool.
6. A database is created in OpenOffice Base with the extension .odb.
7. OpenOffice Base is a relational database management system.

EXERCISE



A. Fill in the blanks.

1. ..... object of the database displays the formatted data from the table.
2. Rows in a database are also called ..... and columns are also called ..... .
3. A database can contain ..... tables.
4. A spreadsheet is a representation of data in the form of ..... and ..... .
5. ..... is the graphical representation of data in a spreadsheet.

B. Match the following.

1. Extension of database	a. Displays total of the values
2. Tuple	b. Forms
3. Database Objects	c. Displays the maximum values
4. Max()	d. Rows in a table
5. Sum()	e. .odb

C. State true or false.

1. Table structure once created cannot be modified later.
2. Text in a spreadsheet is by default left aligned.
3. A cell can have any type of a value in a spreadsheet.

4. We can do calculation on a text value.
5. We cannot delete a record in a table created in OpenOffice Base.

D. Answer the following questions.

1. Differentiate between a formula and function with an example.
2. Explain the Autofill feature with example.
3. Explain the different elements of a database with an example.
4. What is a primary key? What is the advantage of creating a primary key in a table?
5. Explain in short the need for creating a database.



LAB WORK

A. Create a table in a spreadsheet (at least 10 records) with the following columns – Employee Code, Employee Name, Basic Salary, 20% Dearness Allowance, 30% Housing Rent Allowance, 20% Tax Deducted, Gross Salary (Basic + Allowance), Net Salary (Gross-deductions).

B. Use at least three different charts to represent the above data in a graphical form. Try to format the charts created by giving different colours and also swap the rows and columns of the tables on *x* and *y* axis of the graph to see the difference in the appearance.



PROJECT WORK

Visit any shopping mall of your area and try to find out the number of registers maintained to do different jobs for the smooth functioning of the business. After this create a Database with the manual registers converted into tables like stock details, items details sold in a day, details of employees, salary calculation of employees. Add more to this list if you find out a few more different types of registers. Design the structure of each table yourself and add at least five records in each.

QUICK KEY

To open a new OpenOffice file	Ctrl + N
To open an existing OpenOffice file	Ctrl + O
To save an OpenOffice file	Ctrl + S



AVERAGE() It calculates the average of the values given as parameters in the function.

Column The arrangement of relevant type of data vertically in a table forms a field or a column.

Data It is the smallest piece of information.

Database Collection of many relevant tables together will form a database.

IF() It evaluates the condition given as the first parameter. If the condition is true, then the second parameter will be evaluated, otherwise the third parameter will be evaluated.

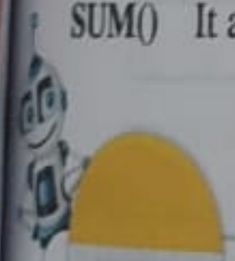
MAX() It returns the highest value from the list of the values given as parameters in the function.

MIN() It returns the lowest value from the list of the values given as parameters in the function.

OpenOffice Base It is a powerful software application used for storing and maintaining the data in the form of rows and columns.

OpenOffice Calc It is part of the OpenOffice suite used as a spreadsheet tool.

SUM() It adds up the values given as parameters in the function.



1. A file in OpenOffice Calc is referred to as a workbook with the default name Untitled 1.
2. A spreadsheet is created in OpenOffice Calc with the extension .ods.
3. In a spreadsheet, you can generate a series of number or text values by using the AutoFill handle.